



101

> Must-Know

Challenging Maths Word Problems

Book

3

Based on current Primary Mathematics Syllabus

- Improves student's ability to solve challenging word problems
- Encourages critical thinking
- Various problem-solving strategies revealed
- Step-by-step solutions provided



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**Challenging
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3

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Name: _____

Class: _____

Preface

101 Must-Know Challenging Maths Word Problems Book 3 presents word problems that test on important concepts so students can learn to **apply general mathematical problem-solving strategies and heuristics confidently**.

What's in this book?

This book comprises word problems often encountered by students in their tests and examinations. The questions are categorized into respective topics in accordance with the current **Primary Mathematics Syllabus**.

Solutions

Detailed step-by-step workings are included in the answer key for every question to show how a problem is solved. **Diagrams and mathematical models** are provided in most solutions to aid students in understanding the problem-solving processes.



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Read about famous mathematicians who contributed to creative mathematics. Develop and hone creative and critical thinking skills through free sample questions.

The Editorial Team

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Question

The sum of 3 numbers is 1305. The first number is 360. The second number is twice the first number. What is the third number?

**1**

Answer: _____

Question

Mary had 129 apples. Tom had 127 more apples than Mary. Linda had 195 more apples than Tom. How many more apples did Linda have than Mary?

**2**

Answer: _____

Question

Jacob had twice as many marbles as Jerry. Kenny had 3 times as many marbles as Jacob. They had 846 marbles altogether. How many more marbles did Kenny have than Jerry?

**3**

Answer: _____

Question

There were 5 times as many strawberries as blueberries in a container. Joel ate half of the number of strawberries and 120 strawberries were left in the container. How many strawberries and blueberries were in the container at first?

Answer: _____

Question

There were 67 more boys than girls in a field. After 18 children had left the field, there were 373 children left. How many boys were there in the beginning?

Answer: _____

Question

Jessica had twice as many sweets as Linda and 25 fewer sweets than Zoe. They had 100 sweets altogether. How many sweets did Zoe have?

Answer: _____

Question**7**

315 children were going to the zoo. Each bus could seat 30 children. What was the least number of buses needed to take all the children to the zoo?

Answer: _____

Question**8**

Mrs Woods bought 12 packs of sweets and gave them to 26 students. When she gave each of them 4 sweets, she had 4 sweets left. How many sweets were there in each pack?

Answer: _____

Question**9**

A shopkeeper sold 254 eggs on Monday and twice as many eggs on Tuesday. He had 150 eggs left. If he had to sell all the eggs and the same number of eggs over 2 days, how many eggs would he need to sell on each day?

Answer: _____

Question10

A pack of sweets was shared equally among 30 boys. 4 boys decided to give their sweets to the rest. As a result, the rest of the boys received 2 more sweets each. How many sweets were there in the pack at first?

Answer: _____

Question11

There were 120 black chairs and 3 times as many white chairs in a room. The chairs were arranged in rows of 8. How many rows of chairs were there?

Answer: _____

Question12

There were 120 yellow marbles and twice as many green marbles. The rest were blue marbles. If there were 74 blue marbles, how many marbles were there altogether?

Answer: _____

Question

Mr Gibb had 567 eggs. He sold 143 eggs and packed the remaining eggs equally into 8 boxes. How many eggs were there in each box?

**13**

Answer: _____

Question

There were 720 orange balloons and twice as many yellow balloons. The rest were green balloons. If there were 2500 balloons altogether, how many green balloons were there?

**14**

Answer: _____

Question

Samuel had 840 sweets. He had 4 times as many sweets as Cheryl. How many sweets did they have altogether?

**15**

Answer: _____

Question16

There were 250 fiction books and 180 non-fiction books on a bookshelf. There were 3 times as many non-fiction books as scientific books. How many books were there altogether?

Answer: _____

Question17

The sum of two numbers is 185. One number is 15 more than the other number. What are the two numbers?

Answer: _____

Question18

Mrs Drew baked 104 apple pies and 216 lemon pies. She packed the two types of pies separately into boxes of 4 each. How many more boxes of lemon pies than apple pies did she have?

Answer: _____

Question**19**

At a dog show, there were twice as many brown dogs as white dogs and 3 times as many grey dogs as white dogs. There were 342 dogs altogether. How many grey dogs were there?

Answer: _____

Question**20**

A total of 2650 people were on a cruise ship. 1942 of them were adults. If there were 20 more boys than girls, how many girls were there?

Answer: _____

Question**21**

When a number is tripled, it is 16 more than 284. What is the number?

Answer: _____

Question22

There are 10 balloons in a pack. Christine buys 9 such packs. She blows them and ties them into bunches of 15. How many bunches of balloons does she have?

Answer: _____

Question23

Andrew and Deon sold a total of 116 charity tickets. Andrew sold 3 times as many charity tickets as Deon. How many charity tickets did Andrew sell?

Answer: _____

Question24

In a car park, there were 198 cars. There were 54 more cars than vans and 36 more motorcycles than vans. How many vehicles were there altogether?

Answer: _____

Question25

Mrs Campbell baked 7 trays of cookies. There were 48 butter cookies and 21 chocolate chip cookies on each tray. How many more butter cookies than chocolate chip cookies did Mrs Campbell bake?

Answer: _____

Question26

There were 24 eggs in a carton. Mr Taylor bought 4 cartons and used 18 eggs. He then packed the remaining eggs into boxes of 6. How many boxes of eggs did he have?

Answer: _____

Question27

At a party, there were 8 groups of 55 children. Another 6 boys and 19 girls joined the party. How many children were there altogether?

Answer: _____

Question28

Mr Smith sold 152 apples on Monday and 168 apples on Tuesday. He had twice as many apples left as the total number of apples sold. How many apples had he left?

Answer: _____

Question29

Calvin had 148 stamps. There were 3 times as many local stamps as foreign stamps. His mother gave him another 52 local stamps. How many local stamps did Calvin have in the end?

Answer: _____

Question30

Ann gave 5 pencils to each of her friends. She then had 2 pencils left. If she had bought 8 boxes of 9 pencils, how many friends did she give the pencils to?

Answer: _____

Question**31**

There were 8 charity tickets in a booklet. Linda sold 16 booklets and Jill sold 4 times as many booklets as her. Mike sold half the number of booklets Linda sold. How many charity tickets did they sell altogether?

Answer: _____

Question**32**

There were 15 more boys than girls in a class. 8 boys wore spectacles and 5 more girls than boys wore spectacles. If all the girls in the class wore spectacles, how many children were there in the class?

Answer: _____

Question**33**

Mr Donald had 210 apples and 220 pears. He sold twice as many apples as pears and had 130 pieces of fruit left. How many apples did he sell?

Answer: _____

Question34

Zoe had 6 times as many sweets as Malcolm. They had a total of 56 sweets. How many sweets must Zoe give to Malcolm so that they would have the same number of sweets?

Answer: _____

Question35

Box A contained 250 cards and box B contained half as many cards as box A. Box C contained 3 times as many cards as box B. Find the total number of cards in the three boxes.

Answer: _____

Question36

Max had 120 red balloons and 210 green balloons. Jeslin had twice and Dawn had 3 times as many balloons as him. How many balloons did they have altogether?

Answer: _____

Question**37**

There were 130 marbles in a box. 80 of them were red and the rest were green and yellow. If there were twice as many red marbles as green marbles, how many yellow marbles were there?

Answer: _____

Question**38**

There are 5 times as many strawberries in box A than in box B. There are 60 more strawberries in box A than in box B. What is the total number of strawberries in both boxes?

Answer: _____

Question**39**

There were 890 pages in a book. Don read 3 times as many pages of the book on Saturday than on Sunday. He then had 6 pages left to read. How many pages did he read on Saturday?

Answer: _____

Question40

There are some red balloons and blue balloons in a pack. Half of the blue balloons is twice the number of red balloons. There are 60 balloons in the pack. How many blue balloons are there?

Answer: _____

Question41

Mrs Morris arranged 25 rows of 7 chairs for a school concert. 157 chairs were occupied during the concert. How many chairs were not occupied?

Answer: _____

Question42

Helen packed 23 boxes of 6 apple pies. She then had 3 apple pies left. How many more boxes of apple pies would she have if she had packed them into boxes of 3 instead?

Answer: _____

Question

A clock cost \$24 and a watch cost \$15 more than the clock. Jeffrey bought 2 clocks and 3 watches. How much money did he spend?

**43**

Answer: _____

Question

A buffet lunch cost \$23.20 for an adult and \$7.50 less for a child. Find the total amount of money that 4 adults and 5 children had to pay for the buffet lunch.

**44**

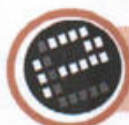
Answer: _____

Question

Joshua had \$150 more than Jason and \$70 less than George. The three boys had a total of \$790. How much did George have?

**45**

Answer: _____



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Question46

Mr Robin spent \$570.80 on a camera and \$155.90 less on a radio. He then had half as much money left as he had in the beginning. How much did he have at first?

Answer: _____

Question47

4 apples and 4 oranges cost \$3. One orange costs 35 cents. How much more does one apple cost than one orange?

Answer: _____

Question48

Mrs William earned \$3500 a month. She spent \$1420 on her children and \$1200 on transport and food. She then saved half of her remaining money. How much would she save in 3 months?

Answer: _____

Question**49**

Adrian had seven 50-cent coins and some 20-cent coins. He spent \$3.70 on a pair of socks and had \$1.80 left. How many 20-cent coins did he have at first?

Answer: _____

Question**50**

A shirt cost \$45.20 and a pair of pants cost \$55.90. Sean wanted to buy 2 such shirts and a pair of pants. He was \$12.45 short of the total amount. How much money did Sean have?

Answer: _____

Question**51**

9 packs of chocolates cost \$72 and 3 packs of sweets cost \$12. Agnes wanted to buy 4 packs of chocolates and 2 packs of sweets. Find the total amount of money that she had to pay for the items.

Answer: _____

Question52

7 people paid \$80 each for a dinner. The amount collected was \$70 short of the total bill. What was the amount of money that each of them should pay?

Answer: _____

Question53

An adult concert ticket cost \$25. A child concert ticket cost \$7 less. Mr and Mrs Lynch took some children to the concert. They paid a total of \$176. How many children did they take to the concert?

Answer: _____

Question54

Joanne had \$18.95. She bought a book that cost \$4.20, a toy that cost \$2.90 and a pair of socks. She had \$9.05 left. How much did she pay for the pair of socks?

Answer: _____

Question**55**

Janet spent half of her salary on food and transport. She then gave \$450 to her brother and saved the remaining amount of \$380. What was Janet's salary?

Answer: _____

Question**56**

2 oranges cost 90 cents and a guava cost 80 cents more than one orange. Jess bought 5 oranges and a guava. She had \$15.75 left. How much did she have at first?

Answer: _____

Question**57**

Mr Anderson bought two similar watches and had \$150 left. He spent 3 times the amount of money he had left on the two watches. How much did each watch cost?

Answer: _____

Question

3 similar books cost \$186. Betty bought 2 books and had \$25 left. How much did she have at first?

**58**

Answer: _____

Question

George earns \$4350 every month. He spends \$350 on transport and \$120 more on food than transport. He then gives \$600 to his wife and saves the rest. How much does he save?

**59**

Answer: _____

Question

7 similar books cost \$105. Benny bought 9 such books and had \$14 left. How much did he have at first?

**60**

Answer: _____

Question61

Mr Fox packed 100 oranges equally into boxes of 5 and sold each box for \$3. Mr Owen packed 100 oranges equally into boxes of 4 and sold each box for \$2. How much more did Mr Fox earn than Mr Owen?

Answer: _____

Question62

Mrs Scotts saved \$564 every month. After half a year, she spent some of her savings. If she had \$1205 left, how much did she spend?

Answer: _____

Question63

A kettle cost \$78. A blender cost \$22 more than the kettle and a toaster cost half as much as the blender. What was the total cost of the 3 electrical items?

Answer: _____

Question**64**

Alex and Bryan shared a cake at a party. Alex ate $\frac{1}{3}$ of the cake while Bryan ate $\frac{1}{9}$ of the cake. What fraction of the cake did they eat altogether?

Answer: _____

Question**65**

Kate bought $\frac{1}{2}$ kg of flour to bake a cake. After baking the cake, she had $\frac{1}{8}$ kg of flour left. How much flour did Kate use?

Answer: _____

Question**66**

Wendy ate $\frac{1}{5}$ of a pizza and Amanda ate $\frac{1}{2}$ of the same pizza. What fraction of the pizza was left?

Answer: _____

Question

George ate $\frac{1}{4}$ of a cake and gave half of the remainder to Amy. What fraction of the cake was left?

**67**

Answer: _____

Question

Bernice bought some sweets. She gave $\frac{1}{4}$ of them to her cousins and $\frac{1}{8}$ of them to her siblings. If she had 70 sweets left, how many sweets did she buy?

**68**

Answer: _____

Question

Alvin, Brenda and Carl shared a cake. Alvin ate $\frac{1}{3}$ of the cake and Brenda ate $\frac{1}{2}$ of the cake. Carl ate the rest of the cake. What fraction of the cake did Carl eat?

**69**

Answer: _____

Question**70**

David ran 1 km 250 m from his house to his school. After that, he cycled 2 km 960 m to the library. What was the total distance he had travelled? (Express your answer in metres.)

Answer: _____

Question**71**

Alan bought 3 m 70 cm of string. He used 1 m 90 cm and cut the remaining length into 6 equal strips. He used 4 strips. What was the length of string left in the end?

Answer: _____

Question**72**

Mrs Cole bought a cloth that was 3 m 50 cm long. She cut it into 5 equal pieces and used 3 such pieces. Find the length of cloth that she had left. (Express your answer in metres and centimetres.)

Answer: _____

Question**73**

Daniel travelled from his home to the beach. He cycled for 2 km 800 m and walked 1 km 950 m less than the distance he cycled. He then took a bus for the rest of the journey. If the distance from his house to the beach was 5 km, what was the distance Daniel travelled by bus? (Express your answer in kilometres and metres.)

Answer: _____

Question**74**

There was 450 g of flour in bag A and 120 g less flour in bag B. Mrs Jackson used 670 g of flour for baking after mixing the two bags together. How much flour was left?

Answer: _____

Question**75**

Mr Clement bought 16 sacks of soil. Each sack contained 9 kg of soil. He repacked the soil equally into bags, each weighing 3 kg. How many bags did he pack?

Answer: _____



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Question76

The mass of a container with a steel ball is 3800 g. When 3 more similar steel balls are added, the mass is 4700 g. What is the mass of the container?

Answer: _____

Question77

Mrs Cox bought 1 kg of flour. She used 480 g and packed the rest equally into 4 bags. How much flour was there in each bag?

Answer: _____

Question78

The total mass of Edward and Ben is 93 kg. The total mass of Edward and Philip is 87 kg. Edward's mass is 36 kg. What is the total mass of Ben and Philip?

Answer: _____

Question**79**

A bottle contained 1 l 250 ml of water. A container had 4 l 180 ml of water. How much more water was there in the container than the bottle? (Express your answer in ml.)

Answer: _____

Question**80**

A flask contains 600 ml of water. A jug contains 150 ml more water than the flask. If a bottle contains 250 ml of water less than the flask, what is the total volume of water in the flask, jug and the bottle?

Answer: _____

Question**81**

Containers A, B and C are half-filled with water. There is 890 ml of water in container A, 120 ml of water in container B and 345 ml of water in container C. If the 3 containers are completely filled, what is the total volume of water that the 3 containers can hold? (Express your answer in litres and millilitres.)

Answer: _____

Question**82**

An empty container was completely filled with water poured from 3 similar jugs. If 5 such jugs contained 400 ml of water, how much water was there in the container?

Answer: _____

Question**83**

Mabel went for a jog last evening. She started jogging from her home at 6.40 pm and returned home 1 hour and 15 minutes later. What time did she return home?

Answer: _____

Question**84**

Leon works in a fast food restaurant. He is paid \$7 an hour. He works 5 hours a day on weekdays and 6 hours a day on weekends. If he works from Monday to Sunday, how much will he earn in a week?

Answer: _____



Carl Fredrich Gauss.

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Question

Mr Watson is 7 times as old as his son this year. His son was 4 years old last year. How old will his son be when Mr Watson is 60 years old?

85

Answer: _____

Question

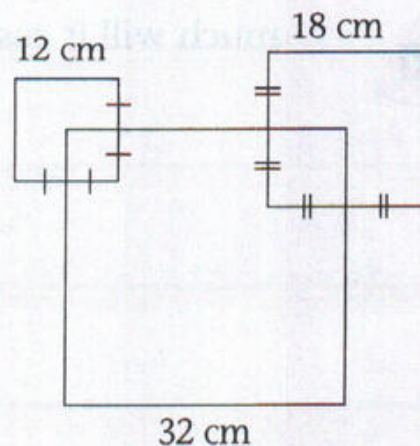
Sue is 3 times as old as Cassandra this year. Cassandra will be 20 years old next year. How old was Sue last year?

86

Answer: _____

Question

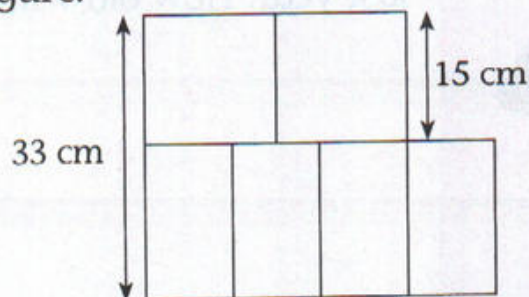
The figure below is made up of 3 squares overlapping each other. Find the perimeter of the figure.

87

Answer: _____

Question**88**

The figure below is made up of 2 similar squares and 4 similar rectangles. Find the perimeter of the figure.



Answer: _____

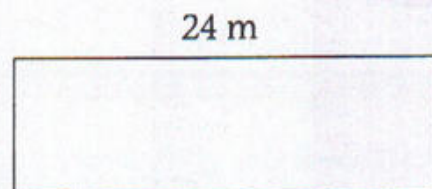
Question**89**

The length of a field is twice its breadth. The perimeter of the field is 162 m. What is the length of the field?

Answer: _____

Question**90**

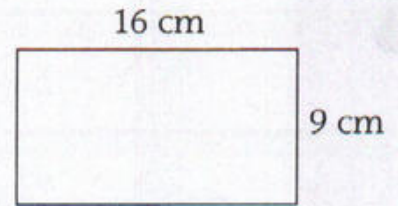
The figure below shows a field. The length of the field is 3 times its breadth. It costs \$8 per metre to put a fence around the field. How much will it cost to fence the entire field?



Answer: _____

Question**91**

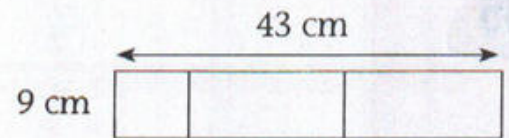
What is the most number of squares with sides of 3 cm that can be cut out from the rectangle shown below?



Answer: _____

Question**92**

The figure below is made up of 2 similar rectangles and a square. Find the total area of the 2 rectangles.



Answer: _____

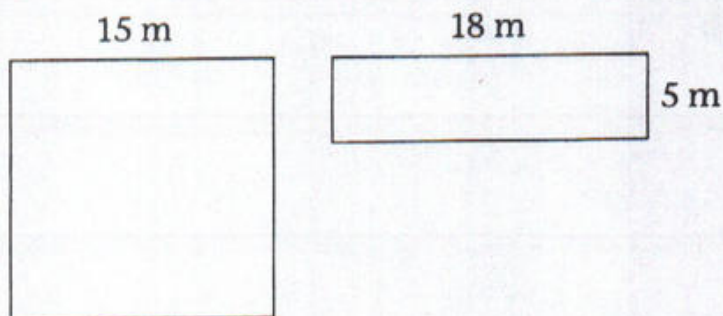
Question**93**

A wire is cut and bent to form 5 similar rectangles. The length of each rectangle is twice its breadth. The length of a rectangle is 24 cm. What is the original length of the wire?

Answer: _____

Question**94**

A square and a rectangle shown below is formed by using half a piece of wire. What is the length of the whole piece of wire?



Answer: _____

Question**95**

A room measuring 12 m by 9 m is to be tiled. If it costs \$6 per square metre to tile the room, what is the total cost of tiling the room?

Answer: _____

Question**96**

After Jeffrey had given 90 sweets to Adeline, she had 270 sweets and he had 140 sweets.

- (a) How many sweets did Jeffrey have at first?
- (b) How many more sweets did Jeffrey have than Adeline at first?

Answers: (a) _____

(b) _____

Question**97**

Sandra had \$45.80. She spent all her money on a photo frame and 2 books. The book cost \$3.80 less than the photo frame.

- (a) How much was the book?
- (b) How much was the photo frame?

Answers: (a) _____

(b) _____

Question**98**

210 children were grouped into 3 teams. If there were 4 times as many boys as girls in each team,

- (a) how many girls were there altogether?
- (b) how many boys were there altogether?

Answers: (a) _____

(b) _____

Question**99**

455 cookies are shared equally among 2 girls and 3 boys. Each girl gets twice as many cookies as each boy.

- (a) How many cookies do the girls get altogether?
- (b) How many cookies do the boys get altogether?

Answers: (a) _____

(b) _____

Question

100

There were 480 apples and oranges in a box. The number of oranges was twice the number of apples in the box.

- (a) How many apples were there?
- (b) How many oranges were there?

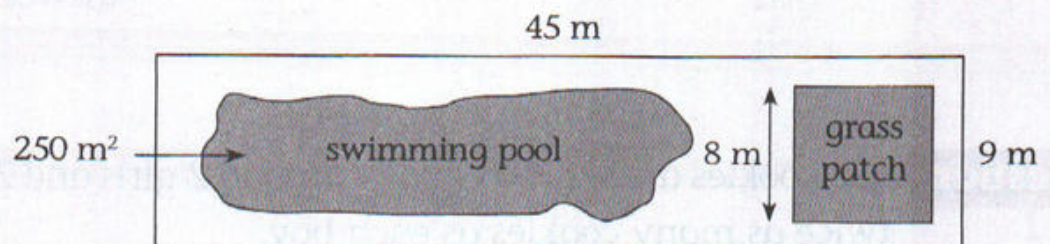
Answers: (a) _____

(b) _____

Question

101

A rectangular plot of land measures 45 m by 9 m. A swimming pool with area of 250 m^2 covers a portion of the land. A square grass patch covers another portion of the land. What is the area of the land that is not covered by the swimming pool and grass patch?



Answer: _____



Pascal's Triangle.
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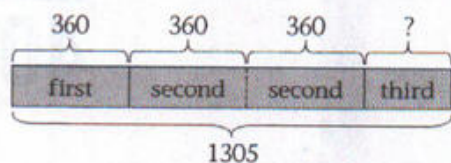


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Solution to Question

1

Step 1 : Draw a model



Step 2 : Find the sum of the first and second numbers

$$3 \times 360 = 1080$$

$$\begin{array}{r} 1 \ 3 \ 6 \ 0 \\ \times \quad 3 \\ \hline 1 \ 0 \ 8 \ 0 \end{array}$$

Step 3 : Find the value of the third number

$$1305 - 1080 = 225$$

$$\begin{array}{r} 2 \ 10 \\ 1 \ 3 \ 0 \ 5 \\ - 1 \ 0 \ 8 \ 0 \\ \hline 2 \ 2 \ 5 \end{array}$$

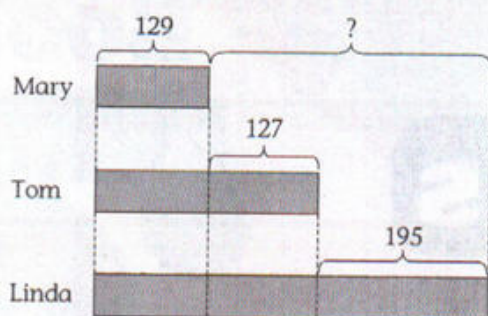
The third number is 225.

Answer: 225

Solution to Question

2

Step 1 : Draw a model



Step 2 : Find how many more apples Linda had than Mary

$$127 + 195 = 322$$

$$\begin{array}{r} 1 \ 1 \\ 1 \ 2 \ 7 \\ + 1 \ 9 \ 5 \\ \hline 3 \ 2 \ 2 \end{array}$$

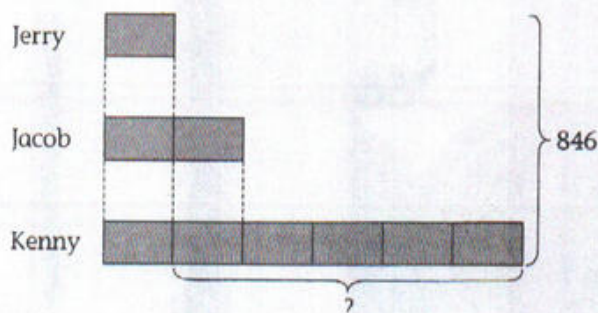
Linda had 322 more apples than Mary.

Answer: 322 more apples

Solution to Question

3

Step 1 : Draw a model



Step 2 : Find the number of marbles Jerry had

$$9 \text{ units} \rightarrow 846$$

$$1 \text{ unit} \rightarrow 846 \div 9 = 94$$

$$\begin{array}{r} 9 \ 4 \\ 9 \overline{) 8 \ 4 \ 6} \\ \underline{- 8 \ 1} \\ 3 \ 6 \\ \underline{- 3 \ 6} \\ 0 \end{array}$$

Step 3 : Compare the number of units in the model to find how many more marbles Kenny had than Jerry

$$5 \text{ units} \rightarrow 5 \times 94 = 470$$

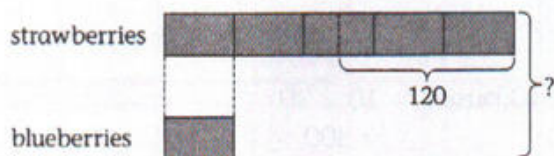
$$\begin{array}{r} 2 \\ 9 \ 4 \\ \times \quad 5 \\ \hline 4 \ 7 \ 0 \end{array}$$

Kenny had 470 more marbles than Jerry.

Answer: 470 marbles

Solution to Question 4

Step 1 : Draw a model



Step 2 : Find the total number of strawberries

$$2 \times 120 = 240$$

$$\begin{array}{r} 120 \\ \times 2 \\ \hline 240 \end{array}$$

Step 3 : Find the total number of blueberries

$$5 \text{ units} \rightarrow 240$$

$$1 \text{ unit} \rightarrow 240 \div 5 = 48$$

$$\begin{array}{r} 48 \\ 5 \overline{) 240} \\ \underline{- 20} \\ 40 \\ \underline{- 40} \\ 0 \end{array}$$

Step 4 : Find the total number of strawberries and blueberries

$$48 + 240 = 288$$

288 strawberries and blueberries were in the container at first.

Answer: 288 strawberries and blueberries

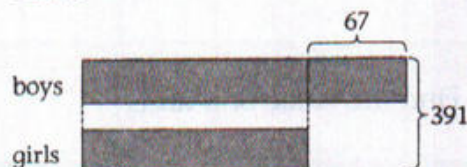
Solution to Question 5

Step 1 : Find the total number of children in the field in the beginning

$$373 + 18 = 391$$

$$\begin{array}{r} 373 \\ + 18 \\ \hline 391 \end{array}$$

Step 2 : Draw a model and find the value of 1 unit



$$2 \text{ units} \rightarrow 391 - 67 = 324$$

$$1 \text{ unit} \rightarrow 324 \div 2 = 162$$

$$\begin{array}{r} 391 \\ - 67 \\ \hline 324 \end{array}$$

$$\begin{array}{r} 162 \\ 2 \overline{) 324} \\ \underline{- 2} \\ 12 \\ \underline{- 12} \\ 0 \end{array}$$

Step 3 : Find the number of boys in the beginning

$$\text{boys} \rightarrow 1 \text{ unit} + 67$$

$$162 + 67 = 229$$

$$\begin{array}{r} 162 \\ + 67 \\ \hline 229 \end{array}$$

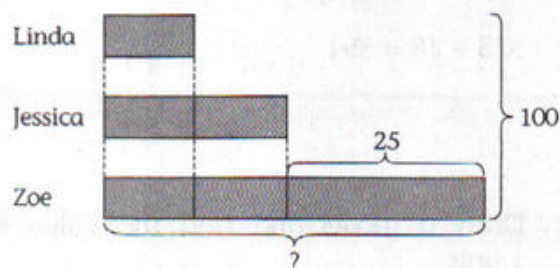
There were 229 boys in the beginning.

Answer: 229 boys

Solution to Question 6

6

Step 1 : Draw a model



Step 2 : Find the value of 5 units

$$100 - 25 = 75$$

$$\begin{array}{r} 9 \ 10 \\ 1 \ 0 \ 0 \\ - \ 2 \ 5 \\ \hline 7 \ 5 \end{array}$$

Step 3 : Find the number of sweets Linda had

$$5 \text{ units} \rightarrow 75$$

$$1 \text{ unit} \rightarrow 75 \div 5 = 15$$

$$\begin{array}{r} 1 \ 5 \\ 5 \overline{) 7 \ 5} \\ - 5 \ 0 \\ \hline 2 \ 5 \\ - 2 \ 5 \\ \hline 0 \end{array}$$

Step 4 : Find the number of sweets Zoe had

$$\text{Zoe} \rightarrow 2 \text{ units} + 25$$

$$2 \text{ units} \rightarrow 2 \times 15 = 30$$

$$30 + 25 = 55$$

Zoe had 55 sweets.

Answer: 55 sweets

Solution to Question 7

7

Step 1 : Find the number of buses required for 300 children

$$1 \text{ bus} \rightarrow 30$$

$$10 \text{ buses} \rightarrow 10 \times 30 = 300$$

$$315 - 300 = 15 \text{ children left}$$

One more bus was needed to take the remaining 15 children.

Step 2 : Find the least number of buses required

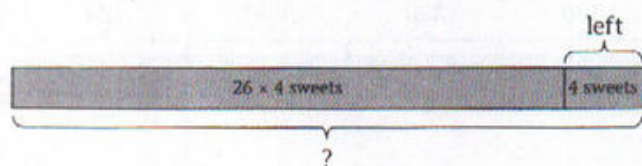
$$10 + 1 = 11$$

The least number of buses needed to take all the children to the zoo was 11.

Answer: 11 buses

Solution to Question 8

Step 1 : Draw a model



Step 2 : Find the number of sweets she gave to her students

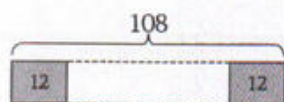
$$26 \times 4 = 104$$

$$\begin{array}{r} 26 \\ \times 4 \\ \hline 104 \end{array}$$

Step 3 : Find the total number of sweets she bought (12 packs)

$$104 + 4 = 108$$

Step 4 : Find the number of sweets in each pack



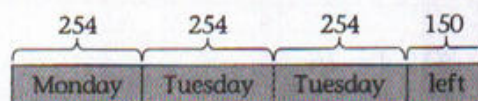
$$108 \div 12 = 9$$

There were 9 sweets in each pack.

Answer: 9 sweets

Solution to Question 9

Step 1 : Draw a model



Step 2 : Find the total number of eggs he had sold

$$3 \times 254 = 762$$

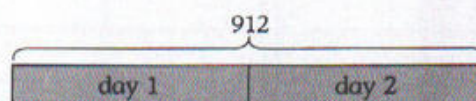
$$\begin{array}{r} 254 \\ \times 3 \\ \hline 762 \end{array}$$

Step 3 : Find the total number of eggs he had at first

$$762 + 150 = 912$$

$$\begin{array}{r} 762 \\ + 150 \\ \hline 912 \end{array}$$

Step 4 : Find the number of eggs he would need to sell each day



$$912 \div 2 = 456$$

$$\begin{array}{r} 456 \\ 2 \overline{) 912} \\ \underline{- 8 } 11 \\ \underline{- 10 } 12 \\ \underline{- 12} 0 \end{array}$$

He would need to sell 456 eggs each day.

Answer: 456 eggs

Solution to Question

10

Step 1 : Find the number of sweets given to rest of the boys

$$30 - 4 = 26$$

$$2 \times 26 = 52$$

$$\begin{array}{r} 1 \\ 26 \\ \times 2 \\ \hline 52 \end{array}$$

Step 2 : Find the number of sweets each of the 4 boys had given away

$$52 \div 4 = 13$$

$$\begin{array}{r} 13 \\ 4 \overline{) 52} \\ - 4 \\ \hline 12 \\ - 12 \\ \hline 0 \end{array}$$

Step 3 : Find the total number of sweets in the pack at first

$$\begin{aligned} 13 \times 30 &= 13 \times 10 \times 3 \\ &= 130 \times 3 \\ &= 390 \end{aligned}$$

$$\begin{array}{r} 130 \\ \times 3 \\ \hline 390 \end{array}$$

There were 390 sweets in the pack at first.

Answer: 390 sweets

Solution to Question

11

Step 1 : Draw a model



Step 2 : Find the total number of chairs

$$4 \times 120 = 480$$

$$\begin{array}{r} 120 \\ \times 4 \\ \hline 480 \end{array}$$

Step 3 : Find the number of rows of chairs

$$480 \div 8 = 60$$

$$\begin{array}{r} 60 \\ 8 \overline{) 480} \\ - 48 \\ \hline 00 \\ - 0 \\ \hline 0 \end{array}$$

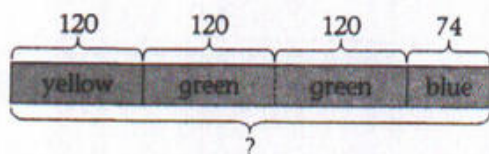
There were 60 rows of chairs.

Answer: 60 rows of chairs

Solution to Question

12

Step 1 : Draw a model



Step 2 : Find the number of green and yellow marbles

$$3 \times 120 = 360$$

$$\begin{array}{r} 120 \\ \times 3 \\ \hline 360 \end{array}$$

Step 3 : Find the total number of marbles

$$360 + 74 = 434$$

$$\begin{array}{r} 360 \\ + 74 \\ \hline 434 \end{array}$$

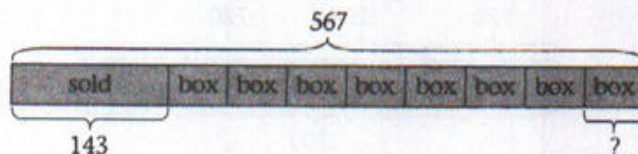
There were 434 marbles altogether.

Answer: 434 marbles

Solution to Question

13

Step 1 : Draw a model



Step 2 : Find the total number of eggs he packed into boxes

$$567 - 143 = 424$$

$$\begin{array}{r} 567 \\ - 143 \\ \hline 424 \end{array}$$

Step 3 : Find the number of eggs in each box

$$424 \div 8 = 53$$

$$\begin{array}{r} 53 \\ 8 \overline{) 424} \\ \underline{- 40} \\ 24 \\ \underline{- 24} \\ 0 \end{array}$$

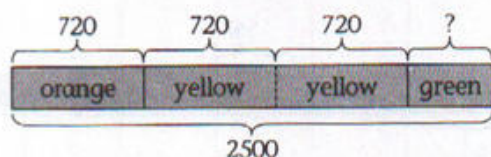
There were 53 eggs in each box.

Answer: 53 eggs

Solution to Question

14

Step 1 : Draw a model



Step 2 : Find the number of orange and yellow balloons

$$3 \times 720 = 2160$$

$$\begin{array}{r} 720 \\ \times 3 \\ \hline 2160 \end{array}$$

Step 3 : Find the number of green balloons

$$2500 - 2160 = 340$$

$$\begin{array}{r} 2500 \\ - 2160 \\ \hline 340 \end{array}$$

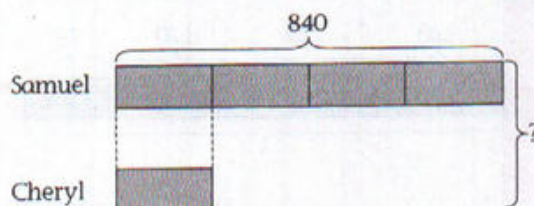
There were 340 green balloons.

Answer: 340 green balloons

Solution to Question

15

Step 1 : Draw a model



Step 2 : Find the number of sweets Cheryl had

$$4 \text{ units} \rightarrow 840$$

$$1 \text{ unit} \rightarrow 840 \div 4 = 210$$

$$\begin{array}{r} 210 \\ 4 \overline{) 840} \\ \underline{- 84} \\ 00 \\ \underline{- 0} \\ 0 \end{array}$$

Step 3 : Find the number of sweets they had altogether

$$840 + 210 = 1050$$

$$\begin{array}{r} 840 \\ + 210 \\ \hline 1050 \end{array}$$

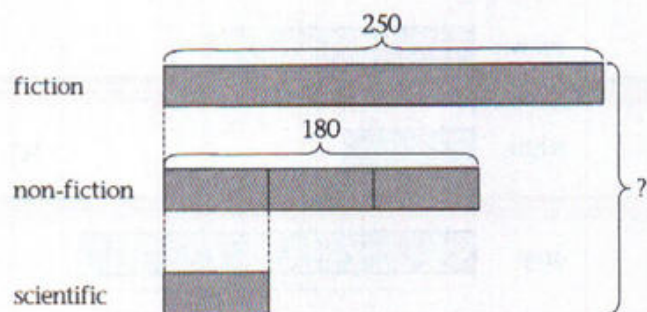
They had 1050 sweets altogether.

Answer: 1050 sweets

Solution to Question

16

Step 1 : Draw a model



Step 2 : Find the number of scientific books

$$3 \text{ units} \rightarrow 180$$

$$1 \text{ unit} \rightarrow 180 \div 3 = 60$$

$$\begin{array}{r} 60 \\ 3 \overline{) 180} \\ \underline{- 180} \\ 00 \\ \underline{- 0} \\ 0 \end{array}$$

Step 3 : Find the total number of books

$$250 + 180 = 430$$

$$430 + 60 = 490$$

$$\begin{array}{r} 1 \\ 250 \\ + 180 \\ \hline 430 \end{array} \quad \begin{array}{r} 430 \\ + 60 \\ \hline 490 \end{array}$$

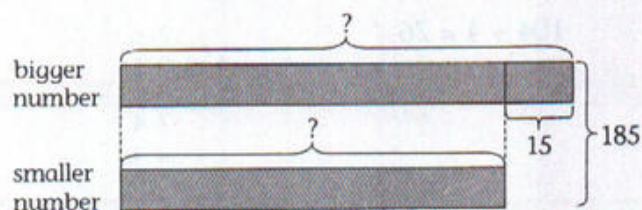
There were 490 books altogether.

Answer: 490 books

Solution to Question

17

Step 1 : Draw a model



Step 2 : Find the value of 1 unit (smaller number)

$$2 \text{ units} \rightarrow 185 - 15 = 170$$

$$1 \text{ unit} \rightarrow 170 \div 2 = 85$$

$$\begin{array}{r} 185 \\ - 15 \\ \hline 170 \\ 2 \overline{) 170} \\ \underline{- 160} \\ 10 \\ \underline{- 10} \\ 0 \end{array}$$

Step 3 : Find the bigger number

$$85 + 15 = 100$$

$$\begin{array}{r} 85 \\ + 15 \\ \hline 100 \end{array}$$

The two numbers are 85 and 100.

Answer: 85 and 100

Solution to Question

18

Step 1 : Find the number of boxes of apple pies

$$104 \div 4 = 26$$

$$\begin{array}{r} 26 \\ 4 \overline{) 104} \\ \underline{- 8} \\ 24 \\ \underline{- 24} \\ 0 \end{array}$$

Step 2 : Find the number of boxes of lemon pies

$$216 \div 4 = 54$$

$$\begin{array}{r} 54 \\ 4 \overline{) 216} \\ \underline{- 20} \\ 16 \\ \underline{- 16} \\ 0 \end{array}$$

Step 3 : Find the difference between the number of boxes of lemon pies and apple pies

$$54 - 26 = 28$$

$$\begin{array}{r} 54 \\ \underline{- 26} \\ 28 \end{array}$$

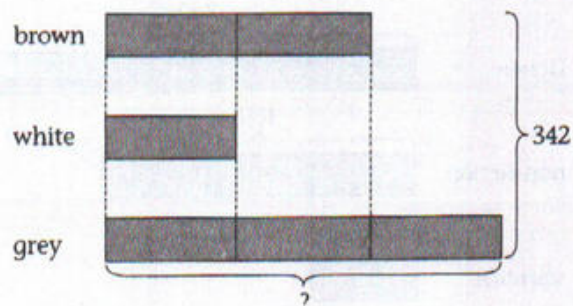
She had 28 more boxes of lemon pies than apple pies.

Answer: 28 more boxes

Solution to Question

19

Step 1 : Draw a model



Step 2 : Find the number of white dogs

$$6 \text{ units} \rightarrow 342$$

$$1 \text{ unit} \rightarrow 342 \div 6 = 57$$

$$\begin{array}{r} 57 \\ 6 \overline{) 342} \\ \underline{- 30} \\ 42 \\ \underline{- 42} \\ 0 \end{array}$$

Step 3 : Find the number of grey dogs

$$3 \text{ units} \rightarrow 3 \times 57 = 171$$

$$\begin{array}{r} 57 \\ \times 3 \\ \hline 171 \end{array}$$

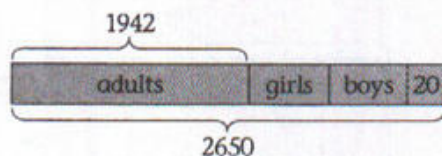
There were 171 grey dogs.

Answer: 171 grey dogs

Solution to Question

20

Step 1 : Draw a model



Step 2 : Find the number of children

$$2650 - 1942 = 708$$

$$\begin{array}{r} 1\ 16\ 4\ 10 \\ 2\ 6\ 5\ 0 \\ - 1\ 9\ 4\ 2 \\ \hline 7\ 0\ 8 \end{array}$$

Step 3 : Find the value of 2 units

$$2 \text{ units} \rightarrow 708 - 20 = 688$$

$$\begin{array}{r} 6\ 10 \\ 3\ 4\ 4 \\ - 2\ 0 \\ \hline 6\ 8\ 8 \end{array}$$

Step 4 : Find the number of girls

$$1 \text{ unit} \rightarrow 688 \div 2 = 344$$

$$\begin{array}{r} 3\ 4\ 4 \\ 2 \overline{) 6\ 8\ 8} \\ \underline{- 6\ 0} \\ 0\ 8 \\ \underline{- 8} \\ 0 \\ \underline{- 0} \\ 0 \end{array}$$

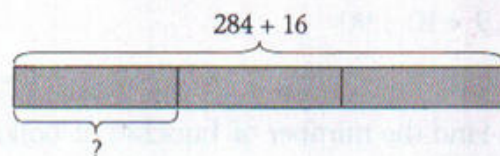
There were 344 girls.

Answer: 344 girls

Solution to Question

21

Step 1 : Draw a model



Step 2 : Find the number when it is tripled

$$284 + 16 = 300$$

$$\begin{array}{r} 1\ 1\ 4 \\ 2\ 8\ 4 \\ + 1\ 6 \\ \hline 3\ 0\ 0 \end{array}$$

Step 3 : Find the number

$$300 \div 3 = 100$$

$$\begin{array}{r} 1\ 0\ 0 \\ 3 \overline{) 3\ 0\ 0} \\ \underline{- 3\ 0} \\ 0 \\ \underline{- 0} \\ 0 \end{array}$$

The number is 100.

Answer: 100

Solution to Question

22

Step 1 : Find the total number of balloons

$$9 \times 10 = 90$$

Step 2 : Find the number of bunches of balloons

$$15 \times 6 = 90$$

$$90 \div 15 = 6$$

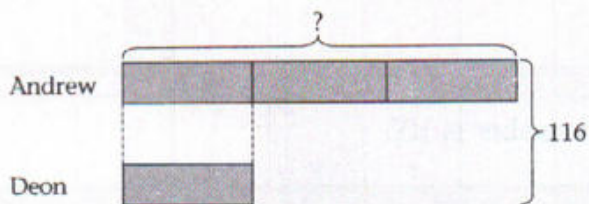
She has 6 bunches of balloons.

Answer: 6 bunches of balloons

Solution to Question

23

Step 1 : Draw a model



Step 2 : Find the number of charity tickets Deon sold

$$4 \text{ units} \rightarrow 116$$

$$1 \text{ unit} \rightarrow 116 \div 4 = 29$$

$$\begin{array}{r} 29 \\ 4 \overline{) 116} \\ \underline{- 8} \\ 36 \\ \underline{- 36} \\ 0 \end{array}$$

Step 3 : Find the number of charity tickets Andrew sold

$$3 \text{ units} \rightarrow 3 \times 29 = 87$$

$$\begin{array}{r} 29 \\ \times 3 \\ \hline 87 \end{array}$$

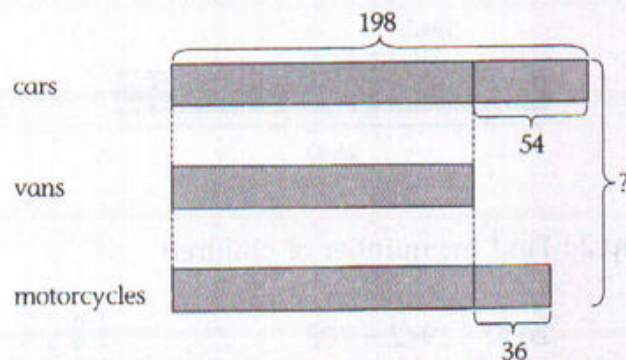
Andrew sold 87 charity tickets.

Answer: 87 charity tickets

Solution to Question

24

Step 1 : Draw a model



Step 2 : Find the number of vans

$$198 - 54 = 144$$

$$\begin{array}{r} 198 \\ - 54 \\ \hline 144 \end{array}$$

Step 3 : Find the number of motorcycles

$$144 + 36 = 180$$

$$\begin{array}{r} 144 \\ + 36 \\ \hline 180 \end{array}$$

Step 4 : Find the total number of vehicles

$$198 + 144 = 342$$

$$342 + 180 = 522$$

$$\begin{array}{r} 198 \\ + 144 \\ \hline 342 \end{array}$$

$$\begin{array}{r} 342 \\ + 180 \\ \hline 522 \end{array}$$

There were 522 vehicles altogether.

Answer: 522 vehicles

Solution to Question

25

Step 1 : Find how many more butter cookies than chocolate chip cookies there are on each tray

$$48 - 21 = 27$$

$$\begin{array}{r} 48 \\ - 21 \\ \hline 27 \end{array}$$

Step 2 : Find how many more butter cookies than chocolate chip cookies there are on 7 trays

$$7 \times 27 = 189$$

$$\begin{array}{r} 27 \\ \times 7 \\ \hline 189 \end{array}$$

Mrs Campbell baked 189 more butter cookies than chocolate chip cookies.

Answer: 189 more butter cookies

Solution to Question

26

Step 1 : Find the total number of eggs he bought

$$4 \times 24 = 96$$

$$\begin{array}{r} 24 \\ \times 4 \\ \hline 96 \end{array}$$

Step 2 : Find the number of remaining eggs

$$96 - 18 = 78$$

$$\begin{array}{r} 96 \\ - 18 \\ \hline 78 \end{array}$$

Step 3 : Find the number of boxes of eggs he had

$$78 \div 6 = 13$$

$$\begin{array}{r} 13 \\ 6 \overline{) 78} \\ - 6 \\ \hline 18 \\ - 18 \\ \hline 0 \end{array}$$

He had 13 boxes of eggs.

Answer: 13 boxes of eggs

Solution to Question

27

Step 1 : Find the total number of children in the 8 groups

$$8 \times 55 = 440$$

$$\begin{array}{r} 55 \\ \times 8 \\ \hline 440 \end{array}$$

Step 2 : Find the total number of children

$$6 + 19 = 25$$

$$\begin{array}{r} 440 \\ + 25 \\ \hline 465 \end{array}$$

$$440 + 25 = 465$$

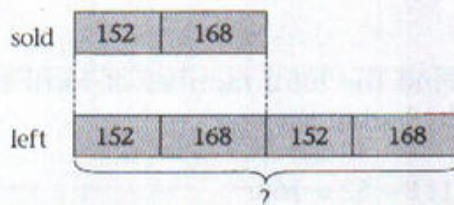
There were 465 children altogether.

Answer: 465 children

Solution to Question

28

Step 1 : Draw a model



Step 2 : Find the total number of apples sold on both days

$$152 + 168 = 320$$

$$\begin{array}{r} 152 \\ + 168 \\ \hline 320 \end{array}$$

Step 3 : Find the number of apples he had left

$$2 \times 320 = 640$$

$$\begin{array}{r} 320 \\ \times 2 \\ \hline 640 \end{array}$$

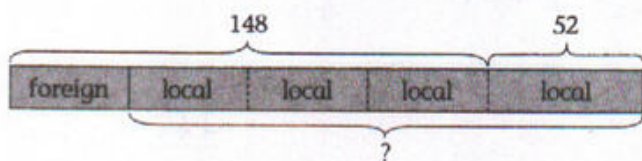
He had 640 apples left.

Answer: 640 apples

Solution to Question

29

Step 1 : Draw a model



Step 2 : Find the number of foreign stamps

4 units $\rightarrow$ 148

1 unit $\rightarrow 148 \div 4 = 37$

$$\begin{array}{r} 37 \\ 4 \overline{) 148} \\ \underline{- 12} \\ 28 \\ \underline{- 28} \\ 0 \end{array}$$

Step 3 : Find the number of local stamps he had at first

$3 \times 37 = 111$

$$\begin{array}{r} 37 \\ \times 3 \\ \hline 111 \end{array}$$

Step 4 : Find the total number of local stamps he had in the end

$111 + 52 = 163$

$$\begin{array}{r} 111 \\ + 52 \\ \hline 163 \end{array}$$

Calvin had 163 local stamps in the end.

Answer: 163 local stamps

Solution to Question

30

Step 1 : Find the total number of pencils

$$8 \times 9 = 72$$

Step 2 : Find the number of pencils she had given to her friends

$$72 - 2 = 70$$

Step 3 : Find the number of friends who were given the pencils

$$70 \div 5 = 14$$

$$\begin{array}{r} 14 \\ 5 \overline{) 70} \\ \underline{- 5} \\ 20 \\ \underline{- 20} \\ 0 \end{array}$$

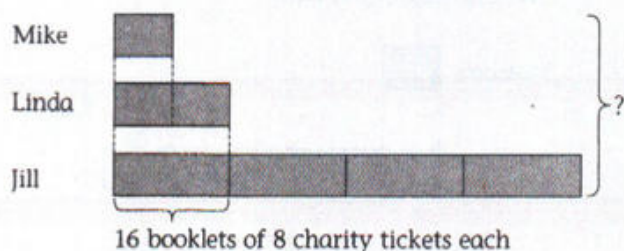
She gave the pencils to 14 friends.

Answer: 14 friends

Solution to Question

31

Step 1 : Draw a model



Step 2 : Find the number of booklets Mike had sold

$$2 \times 8 = 16$$

$$16 \div 2 = 8$$

Step 3 : Find the number of booklets Linda and Jill sold

$$5 \times 16 = 80$$

$$\begin{array}{r} 3 \\ 16 \\ \times 5 \\ \hline 80 \end{array}$$

Step 4 : Find the total number of booklets they had sold

$$80 + 8 = 88$$

Step 5 : Find the total number of charity tickets they had sold

$$8 \times 88 = 704$$

$$\begin{array}{r} 6 \\ 88 \\ \times 8 \\ \hline 704 \end{array}$$

They sold 704 charity tickets altogether.

Answer: 704 charity tickets

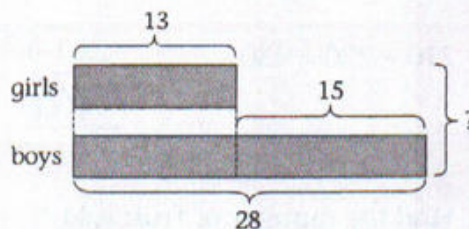
Solution to Question

32

Step 1 : Find the total number of girls in the class

$$8 + 5 = 13$$

Step 2 : Draw a model and find the total number of boys in the class



$$13 + 15 = 28$$

Step 3 : Find the number of children in the class

$$13 + 28 = 41$$

$$\begin{array}{r} 1 \\ 13 \\ + 28 \\ \hline 41 \end{array}$$

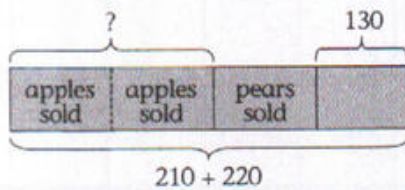
There were 41 children in the class.

Answer: 41 children

Solution to Question

33

Step 1 : Draw a model



Step 2 : Find the total number of fruit

$$210 + 220 = 430$$

$$\begin{array}{r} 210 \\ + 220 \\ \hline 430 \end{array}$$

Step 3 : Find the number of fruit sold

$$430 - 130 = 300$$

$$\begin{array}{r} 430 \\ - 130 \\ \hline 300 \end{array}$$

Step 4 : Find the number of pears sold

$$300 \div 3 = 100$$

$$\begin{array}{r} 100 \\ 3 \overline{) 300} \\ \underline{- 300} \\ 0 \end{array}$$

Step 5 : Find the number of apples sold

$$2 \times 100 = 200$$

$$\begin{array}{r} 100 \\ \times 2 \\ \hline 200 \end{array}$$

He sold 200 apples.

Answer: 200 apples

Solution to Question

34

Step 1 : Draw a model and find the number of sweets Malcolm had



$$56 \div 7 = 8$$

$$7 \times 8 = 56$$

Step 2 : Find the number of sweets each of them should have in order to have the same number of sweets

$$56 \div 2 = 28$$

$$\begin{array}{r} 28 \\ 2 \overline{) 56} \\ \underline{- 40} \\ 16 \\ \underline{- 16} \\ 0 \end{array}$$

Step 3 : Find the number of sweets Zoe must give to Malcolm

$$28 - 8 = 20$$

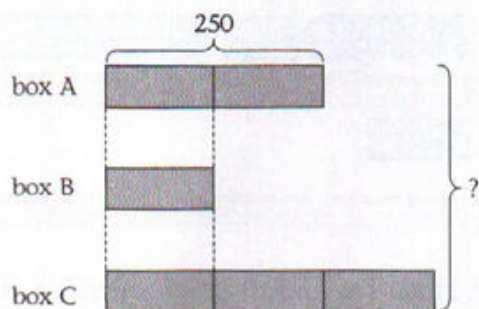
Zoe must give 20 sweets to Malcolm so that they would have the same number of sweets.

Answer: 20 sweets

Solution to Question

35

Step 1 : Draw a model



Step 2 : Find the number of cards in box B

$$2 \text{ units} \rightarrow 250$$

$$1 \text{ unit} \rightarrow 250 \div 2 = 125$$

$$\begin{array}{r} 125 \\ 2 \overline{) 250} \\ \underline{- 200} \\ 50 \\ \underline{- 40} \\ 10 \\ \underline{- 10} \\ 0 \end{array}$$

Step 3 : Find the total number of cards

$$6 \text{ units} \rightarrow 6 \times 125 = 750$$

$$\begin{array}{r} 125 \\ \times 6 \\ \hline 750 \end{array}$$

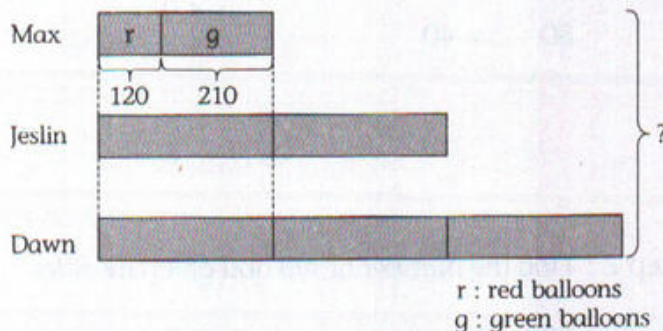
The total number of cards in the three boxes was 750.

Answer: 750 cards

Solution to Question

36

Step 1 : Draw a model



Step 2 : Find the number of balloons Max had

$$120 + 210 = 330$$

$$\begin{array}{r} 120 \\ + 210 \\ \hline 330 \end{array}$$

Step 3 : Find the total number of balloons they had

$$1 \text{ unit} \rightarrow 330$$

$$6 \text{ units} \rightarrow 6 \times 330 = 1980$$

$$\begin{array}{r} 1980 \\ \times 6 \\ \hline 11880 \end{array}$$

They had 1980 balloons altogether.

Answer: 1980 balloons

Solution to Question

37

Step 1 : Find the number of green marbles

$$80 \div 2 = 40$$

$$\begin{array}{r} 40 \\ 2 \overline{) 80} \\ \underline{- 80} \\ 00 \\ \underline{- 0} \\ 0 \end{array}$$

Step 2 : Find the number of red and green marbles

$$80 + 40 = 120$$

$$\begin{array}{r} 80 \\ + 40 \\ \hline 120 \end{array}$$

Step 3 : Find the number of yellow marbles

$$130 - 120 = 10$$

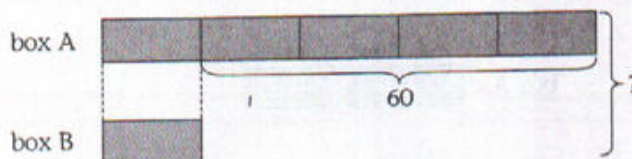
There were 10 yellow marbles.

Answer: 10 yellow marbles

Solution to Question

38

Step 1 : Draw a model



Step 2 : Find the number of strawberries in box B

$$4 \text{ units} \rightarrow 60$$

$$1 \text{ unit} \rightarrow 60 \div 4 = 15$$

$$\begin{array}{r} 15 \\ 4 \overline{) 60} \\ \underline{- 40} \\ 20 \\ \underline{- 20} \\ 0 \end{array}$$

Step 3 : Find the total number of strawberries

$$6 \text{ units} \rightarrow 6 \times 15 = 90$$

$$\begin{array}{r} 90 \\ 6 \times 15 \\ \hline 90 \end{array}$$

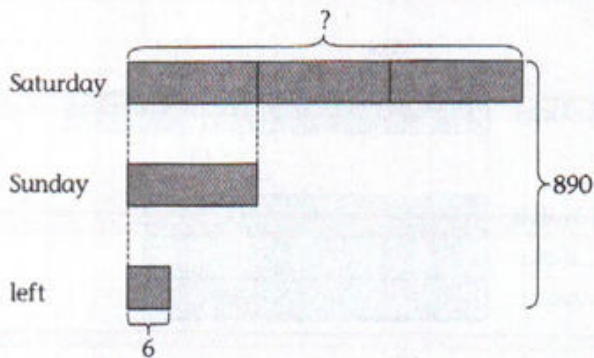
The total number of strawberries in both boxes is 90.

Answer: 90 strawberries

Solution to Question

39

Step 1 : Draw a model



Step 2 : Find the total number of pages he read on Saturday and Sunday

$$890 - 6 = 884$$

$$\begin{array}{r} 890 \\ - 6 \\ \hline 884 \end{array}$$

Step 3 : Find the number of pages he read on Sunday

$$4 \text{ units} \rightarrow 884$$

$$1 \text{ unit} \rightarrow 884 \div 4 = 221$$

$$\begin{array}{r} 221 \\ 4 \overline{) 884} \\ \underline{- 8} \\ 08 \\ \underline{- 8} \\ 4 \\ \underline{- 4} \\ 0 \end{array}$$

Step 4 : Find the number of pages he read on Saturday

$$3 \text{ units} \rightarrow 3 \times 221 = 663$$

$$\begin{array}{r} 221 \\ \times 3 \\ \hline 663 \end{array}$$

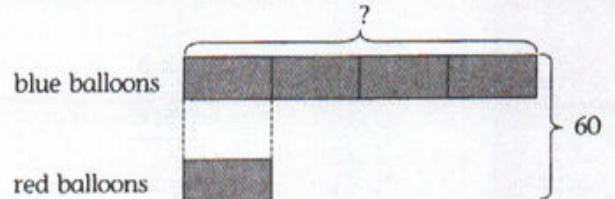
He read 663 pages on Saturday.

Answer: 663 pages

Solution to Question

40

Step 1 : Draw a model



Step 2 : Find the number of red balloons

$$5 \text{ units} \rightarrow 60$$

$$1 \text{ unit} \rightarrow 60 \div 5 = 12$$

$$\begin{array}{r} 12 \\ 5 \overline{) 60} \\ \underline{- 5} \\ 10 \\ \underline{- 10} \\ 0 \end{array}$$

Step 3 : Find the number of blue balloons

$$4 \text{ units} \rightarrow 4 \times 12 = 48$$

$$\begin{array}{r} 12 \\ \times 4 \\ \hline 48 \end{array}$$

There are 48 blue balloons.

Answer: 48 blue balloons

Solution to Question

41

Step 1 : Find the total number of chairs

$$7 \times 25 = 175$$

$$\begin{array}{r} 25 \\ \times 7 \\ \hline 175 \end{array}$$

Step 2 : Find the number of chairs that were not occupied

$$175 - 157 = 18$$

$$\begin{array}{r} 175 \\ - 157 \\ \hline 18 \end{array}$$

18 chairs were not occupied.

Answer: 18 chairs

Solution to Question

42

Step 1 : Find the total number of apple pies

$$6 \times 23 = 138$$

$$138 + 3 = 141$$

$$\begin{array}{r} 23 \\ \times 6 \\ \hline 138 \end{array}$$

Step 2 : Find the number of boxes if 3 apple pies were packed in a box

$$141 \div 3 = 47$$

$$\begin{array}{r} 47 \\ 3 \overline{)141} \\ \underline{-12} \\ 21 \\ \underline{-21} \\ 0 \end{array}$$

Step 3 : Find how many more boxes of 3 apple pies than boxes of 6 apple pies she would have

$$47 - 23 = 24$$

$$\begin{array}{r} 47 \\ - 23 \\ \hline 24 \end{array}$$

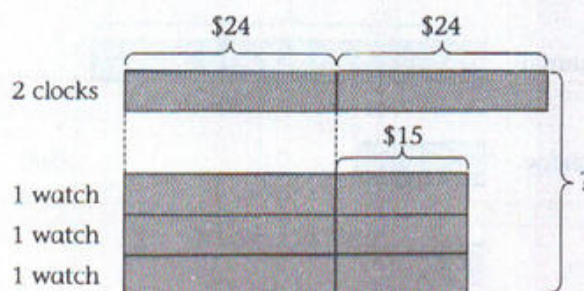
She would have 24 more boxes of apple pies.

Answer: 24 more boxes

Solution to Question

43

Step 1 : Draw a model



Step 2 : Find the cost of 3 watches

$$\$24 + \$15 = \$39$$

$$3 \times \$39 = \$117$$

$$\begin{array}{r} 24 \\ + 15 \\ \hline 39 \end{array} \quad \begin{array}{r} 39 \\ \times 3 \\ \hline 117 \end{array}$$

Step 3 : Find the cost of 2 clocks

$$2 \times \$24 = \$48$$

$$\begin{array}{r} 24 \\ \times 2 \\ \hline 48 \end{array}$$

Step 4 : Find the amount of money he spent

$$\$48 + \$117 = \$165$$

$$\begin{array}{r} 48 \\ + 117 \\ \hline 165 \end{array}$$

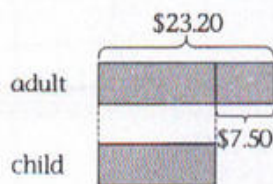
He spent \$165.

Answer: \$165

Solution to Question

44

Step 1 : Draw a model



Step 2 : Find the amount of money that a child had to pay

$$\begin{array}{r} \$23.20 - \$7.50 = \$15.70 \\ \begin{array}{r} 1 \ 12 \ 12 \\ 2 \ 3. \ 2 \ 0 \\ - \quad 7. \ 5 \ 0 \\ \hline 1 \ 5. \ 7 \ 0 \end{array} \end{array}$$

Step 3 : Find the total cost of the buffet lunch for 4 adults

$$\begin{array}{r} 4 \times \$23.20 = \$92.80 \\ \begin{array}{r} 1 \\ 2 \ 3. \ 2 \ 0 \\ \times \quad \quad 4 \\ \hline 9 \ 2. \ 8 \ 0 \end{array} \end{array}$$

Step 4 : Find the total cost of the buffet lunch for 5 children

$$\begin{array}{r} 5 \times \$15.70 = \$78.50 \\ \begin{array}{r} 2 \ 3 \\ 1 \ 5. \ 7 \ 0 \\ \times \quad \quad 5 \\ \hline 7 \ 8. \ 5 \ 0 \end{array} \end{array}$$

Step 5 : Find the total amount of money for the buffet lunch

$$\begin{array}{r} \$92.80 + \$78.50 = \$171.30 \\ \begin{array}{r} 1 \ 1 \\ 9 \ 2. \ 8 \ 0 \\ + \quad 7 \ 8. \ 5 \ 0 \\ \hline 1 \ 7 \ 1. \ 3 \ 0 \end{array} \end{array}$$

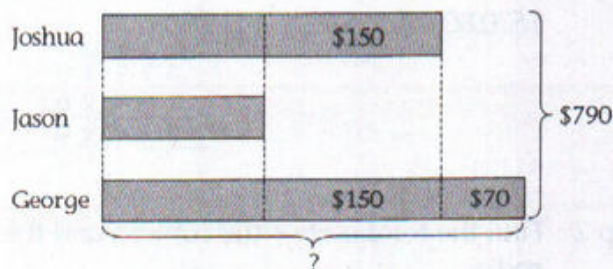
4 adults and 5 children had to pay \$171.30 for the buffet lunch.

Answer: \$171.30

Solution to Question

45

Step 1 : Draw a model



Step 2 : Find the amount of money Joshua and George had more than Jason

$$\begin{array}{r} \$150 + \$70 = \$220 \\ \begin{array}{r} 2 \ 2 \ 0 \\ + \ 1 \ 5 \ 0 \\ \hline 3 \ 7 \ 0 \end{array} \end{array}$$

$$\$220 + \$150 = \$370$$

Step 3 : Find the amount of money Jason had

$$\begin{array}{r} 3 \text{ units} \rightarrow \$790 - \$370 = \$420 \\ \begin{array}{r} 7 \ 9 \ 0 \\ - \ 3 \ 7 \ 0 \\ \hline 4 \ 2 \ 0 \end{array} \end{array}$$

$$\begin{array}{r} 1 \text{ unit} \rightarrow \$420 \div 3 = \$140 \\ \begin{array}{r} 1 \ 4 \ 0 \\ 3 \overline{) 4 \ 2 \ 0} \\ \underline{- \ 3 \ 0} \\ 1 \ 2 \\ \underline{- \ 1 \ 2} \\ 0 \ 0 \\ \underline{- \ 0} \\ 0 \end{array} \end{array}$$

Step 4 : Find the amount of money George had

$$\begin{array}{r} \$140 + \$220 = \$360 \\ \begin{array}{r} 1 \ 4 \ 0 \\ + \ 2 \ 2 \ 0 \\ \hline 3 \ 6 \ 0 \end{array} \end{array}$$

George had \$360.

Answer: \$360

Solution to Question

46

Step 1 : Find the cost of the radio

$$\$570.80 - \$155.90 = \$414.90$$

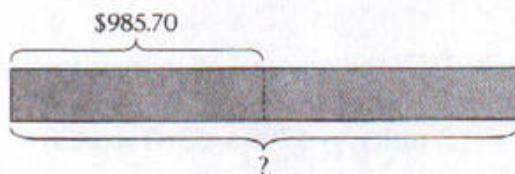
$$\begin{array}{r} ^6 ^9 ^{18} \\ 570.80 \\ - 155.90 \\ \hline 414.90 \end{array}$$

Step 2 : Find the total cost of the camera and the radio

$$\$570.80 + \$414.90 = \$985.70$$

$$\begin{array}{r} ^1 \\ 570.80 \\ + 414.90 \\ \hline 985.70 \end{array}$$

Step 3 : Draw a model and find the amount of money he had at first



$$2 \times \$985.70 = \$1971.40$$

$$\begin{array}{r} ^1 ^1 ^1 \\ 985.70 \\ \times 2 \\ \hline 1971.40 \end{array}$$

He had \$1971.40 at first.

Answer: \$1971.40

Solution to Question

47

Step 1 : Find the cost of 4 oranges

$$35\text{c} \times 4 = 140\text{c}$$

$$140\text{c} = \$1.40$$

$$\$1.00 = 100\text{c}$$

Step 2 : Find the cost of 4 apples

$$\$3.00 - \$1.40 = \$1.60$$

Step 3 : Find the cost of 1 apple

$$\$1.60 \div 4 = \$0.40$$

Step 4 : Find the difference in cost between 1 apple and 1 orange

$$\begin{aligned} \$0.40 - \$0.35 &= \$0.05 \\ &= 5\text{c} \end{aligned}$$

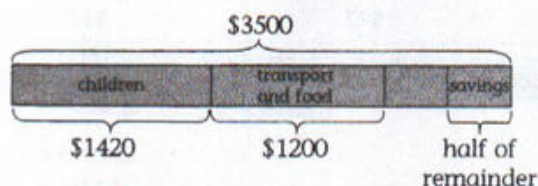
The apple costs 5c more.

Answer: 5c

Solution to Question

48

Step 1 : Draw a model



Step 2 : Find the amount of money she spent on her children, transport and food

$$\begin{array}{r} \$1420 + \$1200 = \$2620 \\ 1420 \\ + 1200 \\ \hline 2620 \end{array}$$

Step 3 : Find the amount of her remaining money

$$\begin{array}{r} \$3500 - \$2620 = \$880 \\ 3500 \\ - 2620 \\ \hline 880 \end{array}$$

Step 4 : Find the amount of money she saved in a month

$$\begin{array}{r} \$880 \div 2 = \$440 \\ 440 \\ 2 \overline{) 880} \\ \underline{- 88} \\ 08 \\ \underline{- 8} \\ 0 \\ \underline{- 0} \\ 0 \end{array}$$

Step 5 : Find the amount of money she would save in 3 months

$$\begin{array}{r} 3 \times \$440 = \$1320 \\ 440 \\ \times 3 \\ \hline 1320 \end{array}$$

She would save \$1320 in 3 months.

Answer: \$1320

Solution to Question

49

Step 1 : Draw a model



Step 2 : Find the amount of money Adrian had

$$\begin{array}{r} \$3.70 + \$1.80 = \$5.50 \\ 3.70 \\ + 1.80 \\ \hline 5.50 \end{array}$$

Step 3 : Find the amount of money he had in 50-cent coins

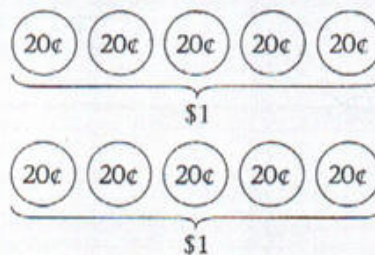
\$1.00 = 100¢

$$\begin{array}{r} 7 \times 50\text{¢} = 350\text{¢} = \$3.50 \\ 50 \\ \times 7 \\ \hline 350 \end{array}$$

Step 4 : Find the amount of money he had in 20-cent coins

$$\begin{array}{r} \$5.50 - \$3.50 = \$2.00 \\ 5.50 \\ - 3.50 \\ \hline 2.00 \end{array}$$

Step 5 : Find the number of 20-cent coins



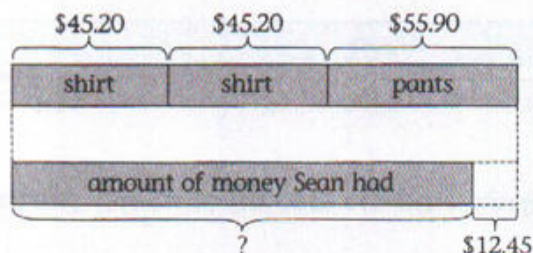
He had ten 20-cent coins at first.

Answer: Ten 20-cent coins

Solution to Question

50

Step 1 : Draw a model



Step 2 : Find the cost of 2 shirts

$$\$45.20 + \$45.20 = \$90.40$$

$$\begin{array}{r} 1 \\ 45.20 \\ + 45.20 \\ \hline 90.40 \end{array}$$

Step 3 : Find the cost of 2 shirts and a pair of pants

$$\$90.40 + \$55.90 = \$146.30$$

$$\begin{array}{r} 1 \\ 90.40 \\ + 55.90 \\ \hline 146.30 \end{array}$$

Step 4 : Find the amount of money he had

$$\$146.30 - \$12.45 = \$133.85$$

$$\begin{array}{r} 5 \quad 12 \quad 10 \\ 146.30 \\ - 12.45 \\ \hline 133.85 \end{array}$$

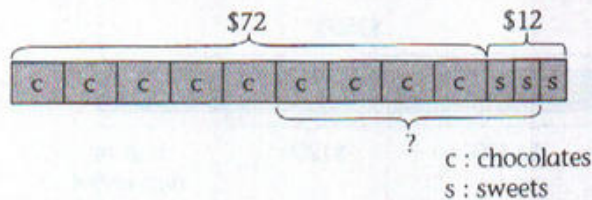
Sean had \$133.85.

Answer: \$133.85

Solution to Question

51

Step 1 : Draw a model



Step 2 : Find the cost of a pack of chocolates

$$9 \times 8 = 72$$

$$\$72 \div 9 = \$8$$

Step 3 : Find the cost of a pack of sweets

$$3 \times 4 = 12$$

$$\$12 \div 3 = \$4$$

Step 4 : Find the cost of 4 packs of chocolates and 2 packs of sweets

$$\text{chocolates} \rightarrow 4 \times \$8 = \$32$$

$$\text{sweets} \rightarrow 2 \times \$4 = \$8$$

Step 5 : Find the total amount she had to pay for the items

$$\$32 + \$8 = \$40$$

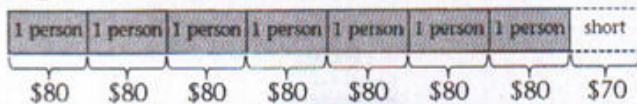
The total amount of money that she had to pay for the items was \$40.

Answer: \$40

Solution to Question

52

Step 1 : Draw a model



Step 2 : Find the amount of money that 7 people had paid

$$7 \times \$80 = \$560$$

$$\begin{array}{r} 80 \\ \times 7 \\ \hline 560 \end{array}$$

Step 3 : Find the cost of the dinner

$$\$560 + \$70 = \$630$$

$$\begin{array}{r} 560 \\ + 70 \\ \hline 630 \end{array}$$

Step 4 : Find the amount of money that each of them should pay

$$\$630 \div 7 = \$90$$

$$\begin{array}{r} 90 \\ 7 \overline{) 630} \\ \underline{- 63} \\ 00 \\ \underline{- 0} \\ 0 \end{array}$$

The amount of money that each of them should pay was \$90.

Answer: \$90

Solution to Question

53

Step 1 : Find the cost of a child concert ticket

$$\$25 - \$7 = \$18$$

Step 2 : Find the cost of 2 adult concert tickets

$$2 \times \$25 = \$50$$

$$\begin{array}{r} 25 \\ \times 2 \\ \hline 50 \end{array}$$

Step 3 : Find the amount paid for all the children

$$\$176 - \$50 = \$126$$

Step 4 : Find the number of children they took to the concert

$$\$126 \div \$18 = 7$$

$$7 \times 18 = 126$$

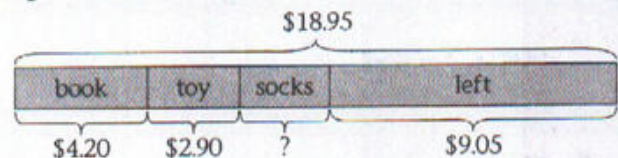
They took 7 children to the concert.

Answer: 7 children

Solution to Question

54

Step 1 : Draw a model



Step 2 : Find the total cost of the book and toy

$$\$4.20 + \$2.90 = \$7.10$$

$$\begin{array}{r} 4.20 \\ + 2.90 \\ \hline 7.10 \end{array}$$

Step 3 : Find the total amount of money spent and left

$$\$7.10 + \$9.05 = \$16.15$$

$$\begin{array}{r} 7.10 \\ + 9.05 \\ \hline 16.15 \end{array}$$

Step 4 : Find the cost of the pair of socks

$$\$18.95 - \$16.15 = \$2.80$$

$$\begin{array}{r} 18.95 \\ - 16.15 \\ \hline 2.80 \end{array}$$

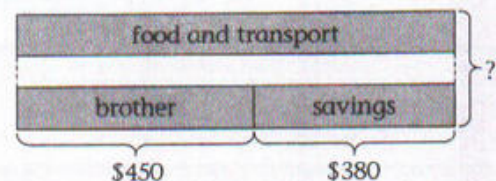
She paid \$2.80 for the pair of socks.

Answer: \$2.80

Solution to Question

55

Step 1 : Draw a model



Step 2 : Find half of her salary

$$\$450 + \$380 = \$830$$

$$\begin{array}{r} 450 \\ + 380 \\ \hline 830 \end{array}$$

Step 3 : Find Janet's salary

$$2 \times \$830 = \$1660$$

$$\begin{array}{r} 830 \\ \times 2 \\ \hline 1660 \end{array}$$

Janet's salary was \$1660.

Answer: \$1660

Solution to Question

56

Step 1 : Find the cost of 1 orange

$$90\text{¢} \div 2 = 45\text{¢}$$

Step 2 : Find the cost of a guava

$$45\text{¢} + 80\text{¢} = 125\text{¢}$$

$$\begin{array}{r} 45 \\ + 80 \\ \hline 125 \end{array}$$

Step 3 : Find the total cost of 5 oranges and 1 guava

$$\text{\$1} = 100\text{¢}$$

$$45\text{¢} \times 5 = 225\text{¢}$$

$$\begin{array}{r} 125\text{¢} + 225\text{¢} = 350\text{¢} \\ = \$3.50 \end{array}$$

$$\begin{array}{r} 45 \\ \times 5 \\ \hline 225 \end{array} \quad \begin{array}{r} 125 \\ + 225 \\ \hline 350 \end{array}$$

Step 4 : Find the amount of money she had at first

$$\begin{array}{r} \$3.50 + \$15.75 = \$19.25 \\ \begin{array}{r} 3.50 \\ + 15.75 \\ \hline 19.25 \end{array} \end{array}$$

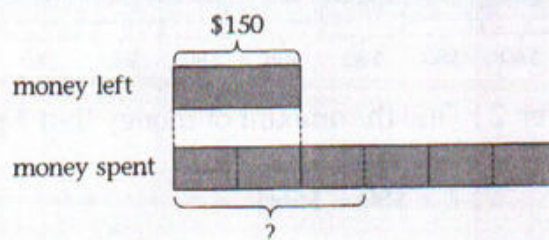
She had \\$19.25 at first.

Answer: \\$19.25

Solution to Question

57

Step 1 : Draw a model



Step 2 : Find the amount of money he had spent

$$3 \times \$150 = \$450$$

$$\begin{array}{r} 150 \\ \times 3 \\ \hline 450 \end{array}$$

Step 3 : Find the cost of each watch

$$\$450 \div 2 = \$225$$

$$\begin{array}{r} 225 \\ 2 \overline{)450} \\ \underline{-40} \\ 50 \\ \underline{-40} \\ 100 \\ \underline{-100} \\ 0 \end{array}$$

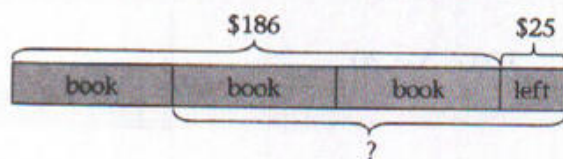
Each watch cost \\$225.

Answer: \\$225

Solution to Question

58

Step 1 : Draw a model



Step 2 : Find the cost of each book

$$\$186 \div 3 = \$62$$

$$\begin{array}{r} 62 \\ 3 \overline{) 186} \\ \underline{- 18} \\ 06 \\ \underline{- 6} \\ 0 \end{array}$$

Step 3 : Find the cost of 2 such books

$$2 \times \$62 = \$124$$

$$\begin{array}{r} 62 \\ \times 2 \\ \hline 124 \end{array}$$

Step 4 : Find the amount of money she had at first

$$\$124 + \$25 = \$149$$

$$\begin{array}{r} 124 \\ + 25 \\ \hline 149 \end{array}$$

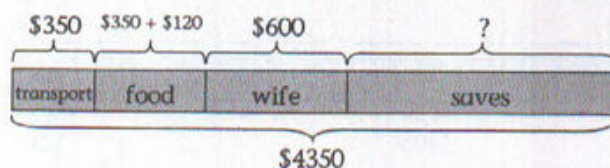
She had \$149 at first.

Answer: \$149

Solution to Question

59

Step 1 : Draw a model



Step 2 : Find the amount of money he spends on food

$$\$350 + \$120 = \$470$$

$$\begin{array}{r} 350 \\ + 120 \\ \hline 470 \end{array}$$

Step 3 : Find the total amount money he spends and gives to his wife

$$\$350 + \$470 = \$820$$

$$\begin{array}{r} 350 \\ + 470 \\ \hline 820 \end{array}$$

$$\$820 + \$600 = \$1420$$

$$\begin{array}{r} 820 \\ + 600 \\ \hline 1420 \end{array}$$

Step 4 : Find the amount of money he saves

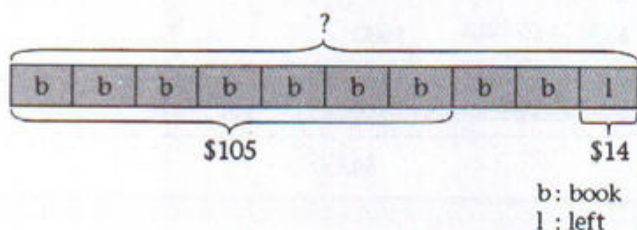
$$\$4350 - \$1420 = \$2930$$

$$\begin{array}{r} 4350 \\ - 1420 \\ \hline 2930 \end{array}$$

He saves \$2930.

Answer: \$2930

Step 1 : Draw a model



Step 2 : Find the cost of each book

$$\$105 \div 7 = \$15$$

$$\begin{array}{r} 15 \\ 7 \overline{) 105} \\ \underline{- 7} \\ 35 \\ \underline{- 35} \\ 0 \end{array}$$

Step 3 : Find the cost of 9 books

$$9 \times \$15 = \$135$$

$$\begin{array}{r} 15 \\ \times 9 \\ \hline 135 \end{array}$$

Step 4 : Find the amount of money he had at first

$$\$135 + \$14 = \$149$$

$$\begin{array}{r} 135 \\ + 14 \\ \hline 149 \end{array}$$

He had \$149 at first.

Answer: \$149

Step 1 : Find the number of boxes of oranges Mr Fox sold

$$100 \div 5 = 20$$

$$\begin{array}{r} 20 \\ 5 \overline{) 100} \\ \underline{- 10} \\ 00 \\ \underline{- 0} \\ 0 \end{array}$$

Step 2 : Find the number of boxes of oranges Mr Owen sold

$$100 \div 4 = 25$$

$$\begin{array}{r} 25 \\ 4 \overline{) 100} \\ \underline{- 8} \\ 20 \\ \underline{- 20} \\ 0 \end{array}$$

Step 3 : Find the amount of money each of them earned

$$\text{Mr Fox} \rightarrow 20 \times \$3 = \$60$$

$$\begin{array}{r} 20 \\ \times 3 \\ \hline 60 \end{array}$$

$$\text{Mr Owen} \rightarrow 25 \times \$2 = \$50$$

$$\begin{array}{r} 25 \\ \times 2 \\ \hline 50 \end{array}$$

Step 4 : Find how much more money Mr Fox earned than Mr Owen

$$\$60 - \$50 = \$10$$

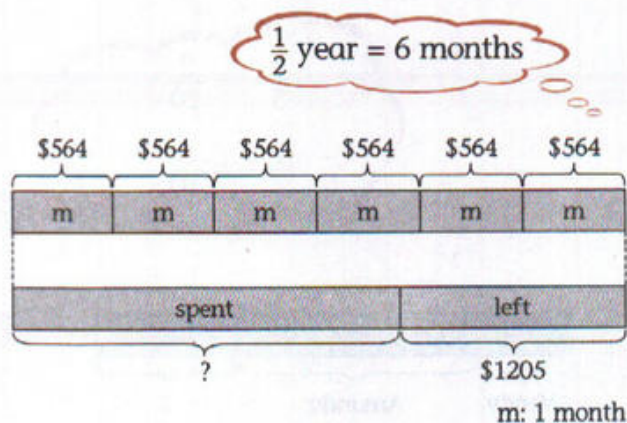
Mr Fox earned \$10 more than Mr Owen.

Answer: \$10 more

Solution to Question

62

Step 1 : Draw a model



Step 2 : Find the total amount of money she had saved in 6 months

$$6 \times \$564 = \$3384$$

$$\begin{array}{r} 3 \ 2 \\ 5 \ 6 \ 4 \\ \times \quad 6 \\ \hline 3 \ 3 \ 8 \ 4 \end{array}$$

Step 3 : Find the amount of money she spent

$$\$3384 - \$1205 = \$2179$$

$$\begin{array}{r} 7 \ 14 \\ 3 \ 3 \ 8 \ 4 \\ - 1 \ 2 \ 0 \ 5 \\ \hline 2 \ 1 \ 7 \ 9 \end{array}$$

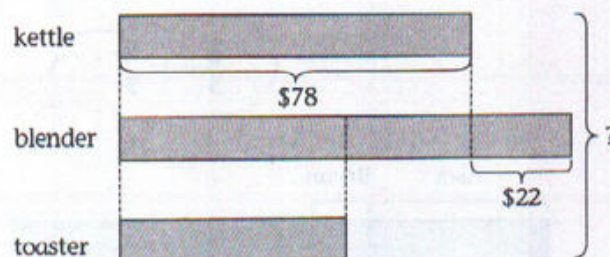
She spent \$2179.

Answer: \$2179

Solution to Question

63

Step 1 : Draw a model



Step 2 : Find the cost of the blender

$$\$78 + \$22 = \$100$$

$$\begin{array}{r} 1 \\ 7 \ 8 \\ + 2 \ 2 \\ \hline 1 \ 0 \ 0 \end{array}$$

Step 3 : Find the cost of the toaster

$$\$100 \div 2 = \$50$$

$$\begin{array}{r} 5 \ 0 \\ 2 \overline{) 1 \ 0 \ 0} \\ - 1 \ 0 \ 0 \\ \hline 0 \ 0 \\ - 0 \\ \hline 0 \end{array}$$

Step 4 : Find the total cost of the 3 electrical items

$$\$78 + \$100 = \$178$$

$$\$178 + \$50 = \$228$$

$$\begin{array}{r} 1 \\ 1 \ 7 \ 8 \\ + 5 \ 0 \\ \hline 2 \ 2 \ 8 \end{array}$$

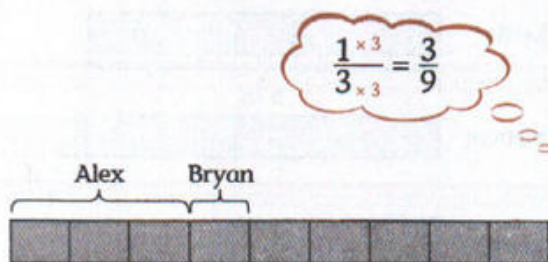
The total cost of the 3 electrical items was \$228.

Answer: \$228

Solution to Question

64

Step 1 : Draw a model



Step 2 : Find how much the boys ate altogether

$$\frac{1}{3} + \frac{1}{9} = \frac{3}{9} + \frac{1}{9} = \frac{4}{9}$$

They ate $\frac{4}{9}$ of the cake altogether.

Answers: $\frac{4}{9}$

Solution to Question

65

Find the amount of flour Kate used

$$\begin{aligned} & \frac{1}{2} \text{ kg} - \frac{1}{8} \text{ kg} \\ &= \frac{4}{8} \text{ kg} - \frac{1}{8} \text{ kg} \\ &= \frac{3}{8} \text{ kg} \end{aligned}$$

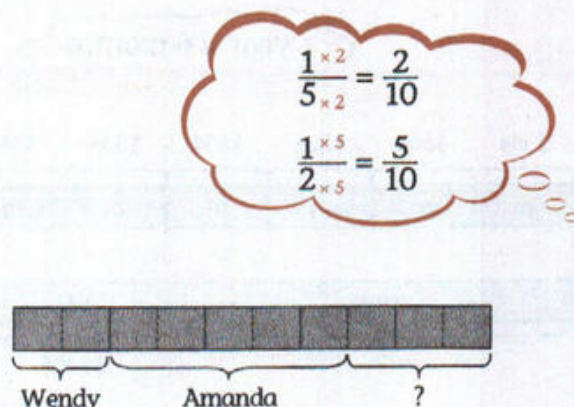
Kate used $\frac{3}{8}$ kg of flour.

Answer: $\frac{3}{8}$ kg

Solution to Question

66

Draw a model



$$\frac{10}{10} - \frac{2}{10} - \frac{5}{10} = \frac{3}{10}$$

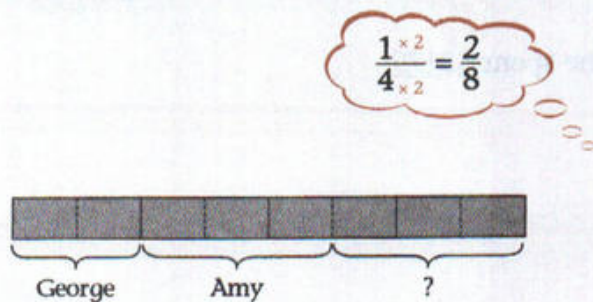
$\frac{3}{10}$ of the pizza was left.

Answer: $\frac{3}{10}$

Solution to Question

67

Draw a model



Total $\rightarrow$ 8 units

Left $\rightarrow 8 - 2 - 3 = 3$ units

Fraction $\rightarrow \frac{3}{8}$

$\frac{3}{8}$ of the cake was left.

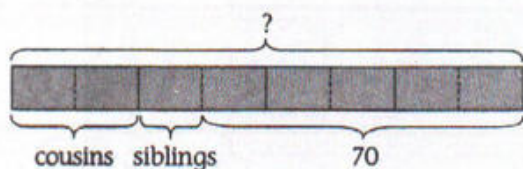
Answer: $\frac{3}{8}$

Solution to Question

68

Step 1 : Draw a model

$$\frac{1 \times 2}{4 \times 2} = \frac{2}{8}$$



Step 2 : Find the number of sweets given to siblings

$$\begin{array}{r} 14 \\ 5 \overline{) 70} \\ \underline{- 50} \\ 20 \\ \underline{- 20} \\ 0 \end{array}$$

5 units $\rightarrow 70$

1 unit $\rightarrow 70 \div 5 = 14$

Step 3 : Find the total number of sweets Bernice bought

$$8 \text{ units} \rightarrow 8 \times 14 = 112$$

$$\begin{array}{r} 112 \\ \times 8 \\ \hline 912 \end{array}$$

She bought 112 sweets.

Answer: 112 sweets

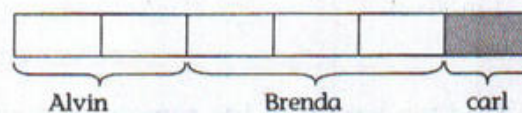
Solution to Question

69

Draw a model

$$\text{Alvin} \rightarrow \frac{1 \times 2}{3 \times 2} = \frac{2}{6}$$

$$\text{Brenda} \rightarrow \frac{1 \times 3}{2 \times 3} = \frac{3}{6}$$



$$\text{Carl} \rightarrow \frac{6}{6} - \frac{2}{6} - \frac{3}{6} = \frac{1}{6}$$

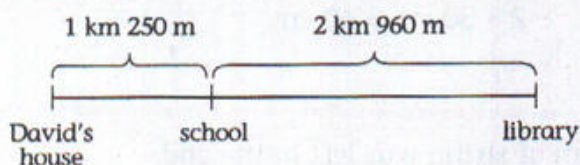
Carl ate $\frac{1}{6}$ of the cake.

Answer: $\frac{1}{6}$

Solution to Question

70

Step 1 : Draw a Time-Line Model



Step 2 : Find the total distance

$$\begin{array}{l} 1 \text{ km} = 1000 \text{ m} \\ 2 \text{ km} = 2000 \text{ m} \end{array}$$

$$1 \text{ km } 250 \text{ m} = 1250 \text{ m}$$

$$2 \text{ km } 960 \text{ m} = 2960 \text{ m}$$

$$1250 \text{ m} + 2960 \text{ m} = 4210 \text{ m}$$

$$\begin{array}{r} 11 \\ 1250 \\ + 2960 \\ \hline 4210 \end{array}$$

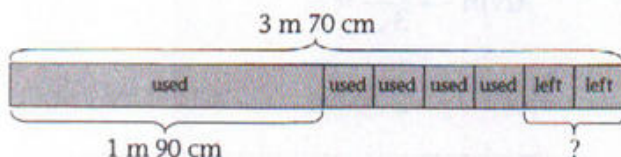
He had travelled a total distance of 4210 m.

Answer: 4210 m

Solution to Question

71

Step 1 : Draw a model



Step 2 : Find the length of the remaining string

1 m = 100 cm

$$\begin{array}{r} 3 \text{ m } 70 \text{ cm} - 1 \text{ m } 90 \text{ cm} \\ = 370 \text{ cm} - 190 \text{ cm} \\ = 180 \text{ cm} \end{array} \quad \begin{array}{r} 2 \text{ } 17 \\ 3 \times 0 \\ - 190 \\ \hline 180 \end{array}$$

Step 3 : Find the length of each strip

$$180 \text{ cm} \div 6 = 30 \text{ cm} \quad \begin{array}{r} 30 \\ 6 \overline{) 180} \\ - 180 \\ \hline 00 \\ - 0 \\ \hline 0 \end{array}$$

Step 4 : Find the length of string he had left

$$2 \times 30 \text{ cm} = 60 \text{ cm} \quad \begin{array}{r} 30 \\ \times 2 \\ \hline 60 \end{array}$$

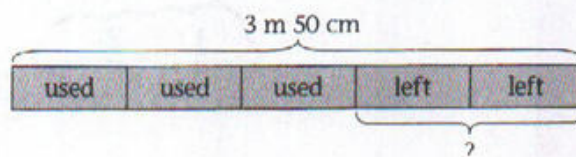
60 cm of string was left in the end.

Answer: 60 cm

Solution to Question

72

Step 1 : Draw a model



Step 2 : Find the length of each piece

1 m = 100 cm

$$\begin{array}{r} 3 \text{ m } 50 \text{ cm} = 350 \text{ cm} \\ 350 \text{ cm} \div 5 = 70 \text{ cm} \end{array} \quad \begin{array}{r} 70 \\ 5 \overline{) 350} \\ - 350 \\ \hline 00 \\ - 0 \\ \hline 0 \end{array}$$

Step 3 : Find the length of cloth that she had left

$$\begin{array}{r} 2 \times 70 \text{ cm} = 140 \text{ cm} \\ = 1 \text{ m } 40 \text{ cm} \end{array} \quad \begin{array}{r} 70 \\ \times 2 \\ \hline 140 \end{array}$$

She had 1 m 40 cm of cloth left.

Answer: 1 m 40 cm

Solution to Question

73

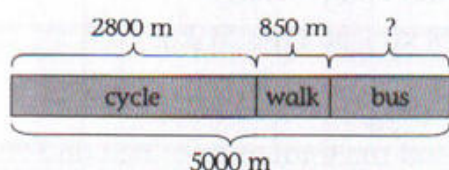
Step 1 : Find the distance he walked

$$1 \text{ km} = 1000 \text{ m}$$

$$\begin{aligned} 2 \text{ km } 800 \text{ m} - 1 \text{ km } 950 \text{ m} \\ = 2800 \text{ m} - 1950 \text{ m} \\ = 850 \text{ m} \end{aligned}$$

$$\begin{array}{r} 1 \text{ } 17 \text{ } 10 \\ 2 \text{ } 8 \text{ } 0 \text{ } 0 \\ - 1 \text{ } 9 \text{ } 5 \text{ } 0 \\ \hline 8 \text{ } 5 \text{ } 0 \end{array}$$

Step 2 : Draw a model



Step 3 : Find the total distance he walked and cycled

$$\begin{aligned} &= 2800 \text{ m} + 850 \text{ m} \\ &= 3650 \text{ m} \end{aligned}$$

$$\begin{array}{r} 1 \text{ } \\ 2 \text{ } 8 \text{ } 0 \text{ } 0 \\ + 8 \text{ } 5 \text{ } 0 \\ \hline 3 \text{ } 6 \text{ } 5 \text{ } 0 \end{array}$$

Step 4 : Find the distance he travelled by bus

$$\begin{aligned} 5000 \text{ m} - 3650 \text{ m} \\ = 1350 \text{ m} \\ = 1 \text{ km } 350 \text{ m} \end{aligned}$$

$$\begin{array}{r} 4 \text{ } 9 \text{ } 10 \\ 5 \text{ } 0 \text{ } 0 \text{ } 0 \\ - 3 \text{ } 6 \text{ } 5 \text{ } 0 \\ \hline 1 \text{ } 3 \text{ } 5 \text{ } 0 \end{array}$$

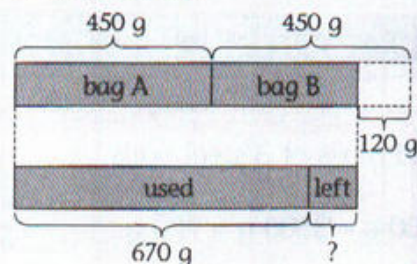
Daniel travelled 1 km 350 m by bus.

Answer: 1 km 350 m

Solution to Question

74

Step 1 : Draw a model



Step 2 : Find the amount of flour in bag B

$$450 \text{ g} - 120 \text{ g} = 330 \text{ g}$$

$$\begin{array}{r} 4 \text{ } 5 \text{ } 0 \\ - 1 \text{ } 2 \text{ } 0 \\ \hline 3 \text{ } 3 \text{ } 0 \end{array}$$

Step 3 : Find the total amount of flour in bags A and B

$$450 \text{ g} + 330 \text{ g} = 780 \text{ g}$$

$$\begin{array}{r} 4 \text{ } 5 \text{ } 0 \\ + 3 \text{ } 3 \text{ } 0 \\ \hline 7 \text{ } 8 \text{ } 0 \end{array}$$

Step 4 : Find the amount of flour left

$$780 \text{ g} - 670 \text{ g} = 110 \text{ g}$$

$$\begin{array}{r} 7 \text{ } 8 \text{ } 0 \\ - 6 \text{ } 7 \text{ } 0 \\ \hline 1 \text{ } 1 \text{ } 0 \end{array}$$

110 g of flour was left.

Answer: 110 g

Solution to Question

75

Step 1 : Find the mass of 16 sacks of soil

$$16 \times 9 \text{ kg} = 144 \text{ kg}$$

$$\begin{array}{r} 5 \text{ } \\ 1 \text{ } 6 \\ \times 9 \\ \hline 1 \text{ } 4 \text{ } 4 \end{array}$$

Step 2 : Find the number of bags he used

$$144 \text{ kg} \div 3 \text{ kg} = 48$$

$$\begin{array}{r} 4 \text{ } 8 \\ 3 \overline{) 1 \text{ } 4 \text{ } 4} \\ - 1 \text{ } 2 \text{ } 4 \\ \hline 2 \text{ } 4 \\ - 2 \text{ } 4 \\ \hline 0 \end{array}$$

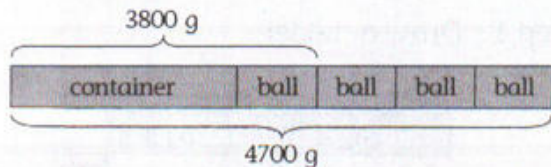
He packed 48 bags.

Answer: 48 bags

Solution to Question

76

Step 1 : Draw a model



Step 2 : Find mass of 3 steel balls

$$4700 \text{ g} - 3800 \text{ g} = 900 \text{ g}$$

$$\begin{array}{r} 3 \text{ } 17 \\ 4 \text{ } 7 \text{ } 0 \text{ } 0 \\ - 3 \text{ } 8 \text{ } 0 \text{ } 0 \\ \hline 9 \text{ } 0 \text{ } 0 \end{array}$$

Step 3 : Find the mass of one steel ball

$$900 \text{ g} \div 3 = 300 \text{ g}$$

$$\begin{array}{r} 3 \text{ } 0 \text{ } 0 \\ 3 \overline{) 9 \text{ } 0 \text{ } 0} \\ \underline{- 9 \text{ } 0} \\ 0 \text{ } 0 \\ \underline{- 0 \text{ } 0} \\ 0 \text{ } 0 \\ \underline{- 0} \\ 0 \end{array}$$

Step 4 : Find the mass of the container

$$3800 \text{ g} - 300 \text{ g} = 3500 \text{ g}$$

$$\begin{array}{r} 3 \text{ } 8 \text{ } 0 \text{ } 0 \\ - 3 \text{ } 0 \text{ } 0 \\ \hline 3 \text{ } 5 \text{ } 0 \text{ } 0 \end{array}$$

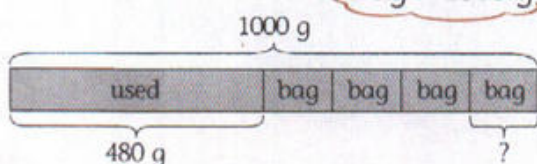
The mass of the container is 3500 g.

Answer: 3500 g

Solution to Question

77

Step 1 : Draw a model



Step 2 : Find the amount of flour packed in 4 bags

$$1000 \text{ g} - 480 \text{ g} = 520 \text{ g}$$

$$\begin{array}{r} 0 \text{ } 9 \text{ } 10 \\ 1 \text{ } 0 \text{ } 0 \text{ } 0 \\ - 4 \text{ } 8 \text{ } 0 \\ \hline 5 \text{ } 2 \text{ } 0 \end{array}$$

Step 3 : Find the amount of flour in each bag

$$520 \text{ g} \div 4 = 130 \text{ g}$$

$$\begin{array}{r} 1 \text{ } 3 \text{ } 0 \\ 4 \overline{) 5 \text{ } 2 \text{ } 0} \\ \underline{- 4 \text{ } 0} \\ 1 \text{ } 2 \\ \underline{- 1 \text{ } 2} \\ 0 \text{ } 0 \\ \underline{- 0} \\ 0 \end{array}$$

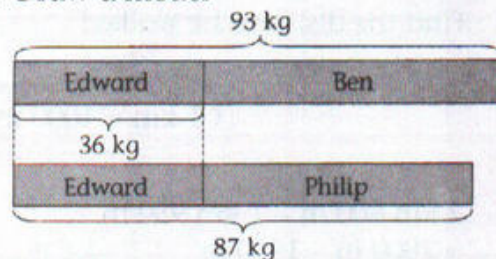
There was 130 g of flour in each bag.

Answer: 130 g

Solution to Question

78

Step 1 : Draw a model



Step 2 : Find Ben's mass

$$93 \text{ kg} - 36 \text{ kg} = 57 \text{ kg}$$

$$\begin{array}{r} 8 \text{ } 13 \\ 9 \text{ } 3 \\ - 3 \text{ } 6 \\ \hline 5 \text{ } 7 \end{array}$$

Step 3 : Find Philip's mass

$$87 \text{ kg} - 36 \text{ kg} = 51 \text{ kg}$$

$$\begin{array}{r} 8 \text{ } 7 \\ - 3 \text{ } 6 \\ \hline 5 \text{ } 1 \end{array}$$

Step 4 : Find the total mass of Ben and Philip

$$57 \text{ kg} + 51 \text{ kg} = 108 \text{ kg}$$

$$\begin{array}{r} 5 \text{ } 7 \\ + 5 \text{ } 1 \\ \hline 1 \text{ } 0 \text{ } 8 \end{array}$$

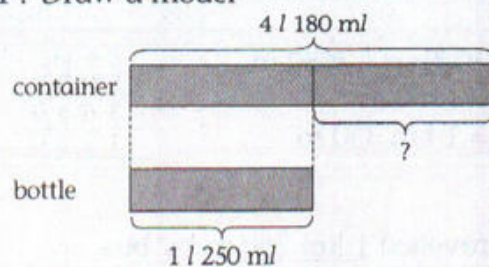
The total mass of Ben and Philip is 108 kg.

Answer: 108 kg

Solution to Question

79

Step 1 : Draw a model



Step 2 : Convert l to ml

$$4 \text{ l } 180 \text{ ml} = 4180 \text{ ml}$$

$$1 \text{ l } 250 \text{ ml} = 1250 \text{ ml}$$

Step 3 : Find the difference in volume

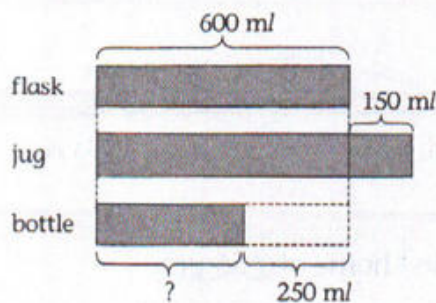
$$4180 \text{ ml} - 1250 \text{ ml} = 2930 \text{ ml}$$

$$\begin{array}{r} 3 \text{ } 11 \\ 4 \text{ } 1 \text{ } 8 \text{ } 0 \\ - 1 \text{ } 2 \text{ } 5 \text{ } 0 \\ \hline 2 \text{ } 9 \text{ } 3 \text{ } 0 \end{array}$$

There was 2930 ml more water in the container than the bottle.

Answer: 2930 ml

Step 1 : Draw a model



Step 2 : Find the volume of the jug

$$600 \text{ ml} + 150 \text{ ml} = 750 \text{ ml}$$

$$\begin{array}{r} 600 \\ + 150 \\ \hline 750 \end{array}$$

Step 3 : Find the volume of the bottle

$$600 \text{ ml} - 250 \text{ ml} = 350 \text{ ml}$$

$$\begin{array}{r} 600 \\ - 250 \\ \hline 350 \end{array}$$

Step 4 : Find the total volume of water in the flask, jug and bottle

$$600 \text{ ml} + 750 \text{ ml} = 1350 \text{ ml}$$

$$\begin{array}{r} 600 \\ + 750 \\ \hline 1350 \end{array}$$

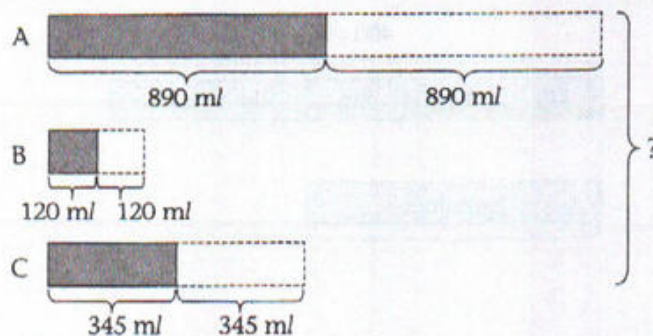
$$1350 \text{ ml} + 350 \text{ ml} = 1700 \text{ ml}$$

$$\begin{array}{r} 1350 \\ + 350 \\ \hline 1700 \end{array}$$

The total volume of water is 1700 ml.

Answers: 1700 ml

Step 1 : Draw a model



Step 2 : Find the total volume of water in the 3 containers when they are half-filled with water

$$890 \text{ ml} + 120 \text{ ml} = 1010 \text{ ml}$$

$$\begin{array}{r} 890 \\ + 120 \\ \hline 1010 \end{array}$$

$$1010 \text{ ml} + 345 \text{ ml} = 1355 \text{ ml}$$

$$\begin{array}{r} 1010 \\ + 345 \\ \hline 1355 \end{array}$$

Step 3 : Find the total volume of water that the 3 containers can hold

$$1 \text{ l} = 1000 \text{ ml}$$

$$2 \times 1355 \text{ ml} = 2710 \text{ ml}$$

$$\begin{array}{r} 1355 \\ \times 2 \\ \hline 2710 \end{array}$$

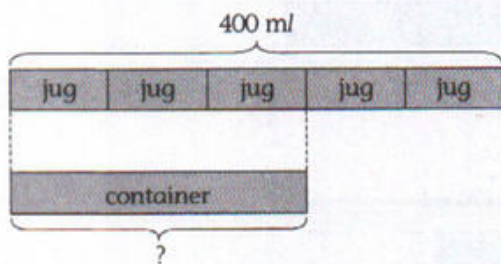
The 3 containers can hold 2 l 710 ml in total.

Answer: 2710 ml

Solution to Question

82

Step 1 : Draw a model



Step 2 : Find the capacity of each jug

$$400 \text{ ml} \div 5 = 80 \text{ ml}$$

$$\begin{array}{r} 80 \\ 5 \overline{) 400} \\ \underline{- 400} \\ 0 \end{array}$$

Step 3 : Find the volume of water in the container

$$3 \times 80 \text{ ml} = 240 \text{ ml}$$

$$\begin{array}{r} 80 \\ \times 3 \\ \hline 240 \end{array}$$

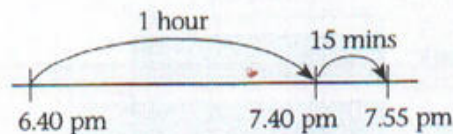
There was 240 ml of water in the container.

Answer: 240 ml

Solution to Question

83

Draw a Time-Line model



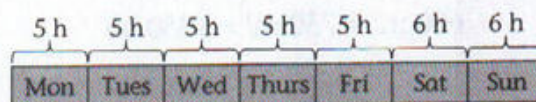
She returned home at 7.55 pm.

Answer: 7.55 pm

Solution to Question

84

Step 1 : Draw a model



Step 2 : Find the total number of working hours in one week

$$5 \text{ weekdays} \rightarrow 5 \times 5 = 25 \text{ hours}$$

$$2 \text{ weekends} \rightarrow 2 \times 6 = 12 \text{ hours}$$

$$\text{Total} \rightarrow 25 + 12 = 37 \text{ hours}$$

Step 3 : Find the total amount Leon will earn in a week

$$37 \times \$7 = \$259$$

$$\begin{array}{r} 4 \\ 37 \\ \times 7 \\ \hline 259 \end{array}$$

He will earn \$259 in a week.

Answer: \$259

Solution to Question

85

Step 1 : Find his son's present age

$$4 + 1 = 5$$

Step 2 : Find Mr Watson's present age

$$7 \times 5 = 35$$

Step 3 : Find the difference in age between Mr Watson and his son

$$35 - 5 = 30$$

Step 4 : Find his son's age when Mr Watson turns 60 years old

$$60 - 30 = 30$$

$$\begin{array}{r} 60 \\ - 30 \\ \hline 30 \end{array}$$

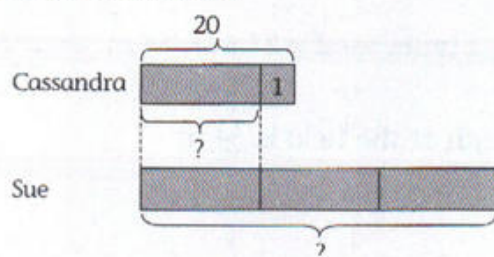
His son will be 30 years old.

Answer: 30 years old

Solution to Question

86

Step 1 : Draw a model



Step 2 : Find Cassandra's present age

$$20 - 1 = 19$$

Step 3 : Find Sue's present age

$$\begin{array}{r} 19 \\ \times 3 \\ \hline 57 \end{array}$$

Step 4 : Find Sue's age last year

$$57 - 1 = 56$$

Sue was 56 years old last year.

Answer: 56 years old

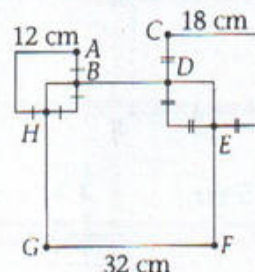
Solution to Question

87

Step 1 : Find the lengths AB and CD

$$AB = 12 \text{ cm} \div 2 = 6 \text{ cm}$$

$$CD = 18 \text{ cm} \div 2 = 9 \text{ cm}$$



Step 2 : Find the lengths EF and GH

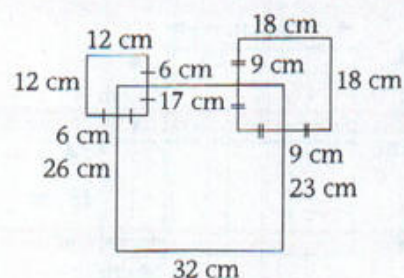
$$EF = 32 \text{ cm} - 9 \text{ cm} = 23 \text{ cm}$$

$$GH = 32 \text{ cm} - 6 \text{ cm} = 26 \text{ cm}$$

Step 3 : Find the length BD

$$BD = 32 \text{ cm} - 6 \text{ cm} - 9 \text{ cm} = 17 \text{ cm}$$

Step 4 : Find the perimeter of the figure

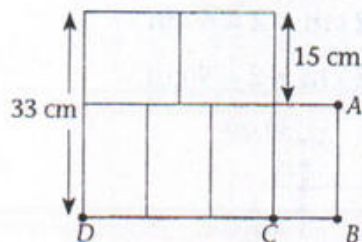


$$12 \text{ cm} + 6 \text{ cm} + 17 \text{ cm} + 9 \text{ cm} + 18 \text{ cm} + 23 \text{ cm} + 26 \text{ cm} + 12 \text{ cm} = 188 \text{ cm}$$

The perimeter of the figure is 188 cm.

Answer: 188 cm

Step 1 : Find the length AB



$$AB = 33 \text{ cm} - 15 \text{ cm} = 18 \text{ cm}$$

$$\begin{array}{r} 2 \ 13 \\ 3 \ 3 \\ - 1 \ 5 \\ \hline 1 \ 8 \end{array}$$

Step 2 : Find the lengths CD and BC

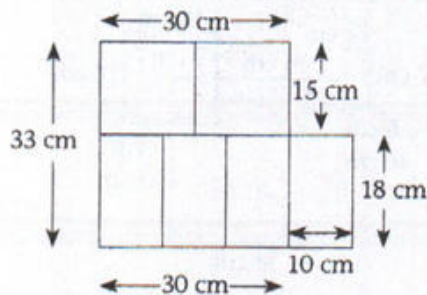
$$CD = 2 \times 15 \text{ cm} = 30 \text{ cm}$$

$$\begin{array}{r} 1 \ 5 \\ \times 2 \\ \hline 3 \ 0 \end{array}$$

$$3 \times 10 = 30$$

$$BC = 30 \text{ cm} \div 3 = 10 \text{ cm}$$

Step 3 : Find the perimeter of the figure



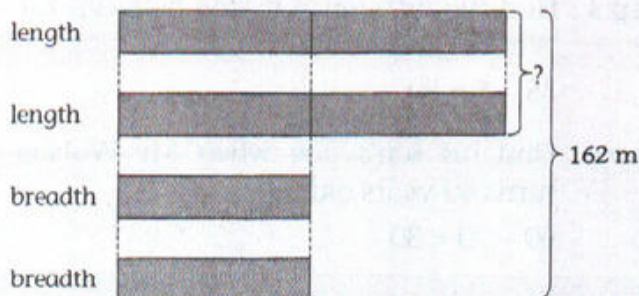
$$30 \text{ cm} + 15 \text{ cm} + 10 \text{ cm} + 18 \text{ cm} + 10 \text{ cm} + 30 \text{ cm} + 33 \text{ cm} = 146 \text{ cm}$$

The perimeter of the figure is 146 cm.

Answer: 146 cm

Step 1 : Draw a model

Perimeter:
2 lengths + 2 breadths



Step 2 : Find the breadth of the field

$$6 \text{ units} \rightarrow 162 \text{ m}$$

$$1 \text{ unit} \rightarrow 162 \text{ m} \div 6 = 27 \text{ m}$$

$$\begin{array}{r} 2 \ 7 \\ 6 \overline{) 1 \ 6 \ 2} \\ - 1 \ 2 \downarrow \\ \hline 4 \ 2 \\ - 4 \ 2 \\ \hline 0 \end{array}$$

Step 3 : Find the length of the field

$$2 \text{ units} \rightarrow 2 \times 27 \text{ m} = 54 \text{ m}$$

$$\begin{array}{r} 1 \ 2 \ 7 \\ \times 2 \\ \hline 5 \ 4 \end{array}$$

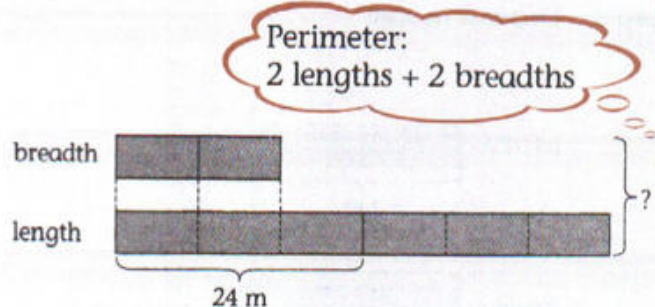
The length of the field is 54 m.

Answer: 54 m

Solution to Question

90

Step 1 : Draw a model



Step 2 : Find the breadth of the field

$$3 \text{ units} \rightarrow 24 \text{ m}$$

$$1 \text{ unit} \rightarrow 24 \text{ m} \div 3 = 8 \text{ m}$$

$$3 \times 8 = 24$$

Step 3 : Find the perimeter of the field

$$\begin{aligned} &\text{length} + \text{breadth} + \text{length} + \text{breadth} \\ &= 24 \text{ m} + 8 \text{ m} + 24 \text{ m} + 8 \text{ m} \\ &= 32 \text{ m} + 32 \text{ m} \\ &= 64 \text{ m} \end{aligned}$$

Step 4 : Find the cost to fence the entire field

$$64 \text{ m} \times \$8 = \$512$$

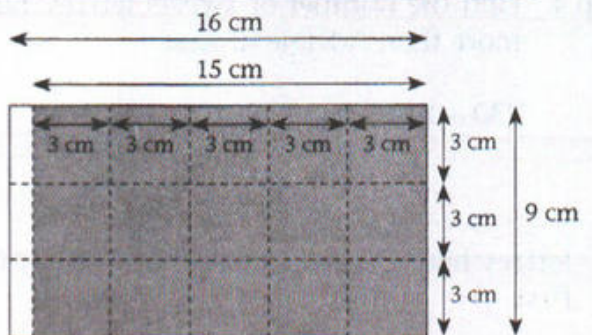
$$\begin{array}{r} 34 \\ \times 8 \\ \hline 512 \end{array}$$

It will cost \$512 to fence the entire field.

Answer: \$512

Solution to Question

91



$$1 \text{ square} = 3 \times 3$$

3 rows of squares

Each row $\rightarrow$ 5 squares (maximum)

$$3 \times 5 = 15$$

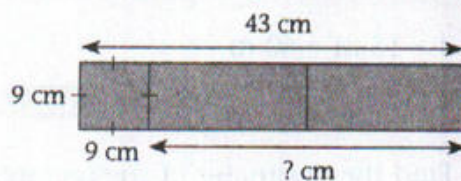
15 squares can be cut out from the rectangle.

Answer: 15 squares

Solution to Question

92

Step 1 : Find the total length of two rectangles



$$43 \text{ cm} - 9 \text{ cm} = 34 \text{ cm}$$

Step 2 : Find the total area of the 2 rectangles

$$34 \text{ cm} \times 9 \text{ cm} = 306 \text{ cm}^2$$

$$\begin{array}{r} 34 \\ \times 9 \\ \hline 306 \end{array}$$

The total area of 2 rectangles is 306 cm².

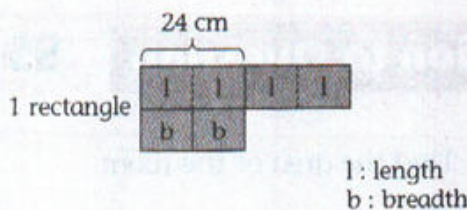
Answer: 306 cm²

Solution to Question

93

Step 1 : Draw a model

Perimeter:
2 lengths + 2 breadths



Step 2 : Find the perimeter of each rectangle

$$3 \times 24 \text{ cm} = 72 \text{ cm}$$

$$\begin{array}{r} 24 \\ \times 3 \\ \hline 72 \end{array}$$

Step 3 : Find the length of the wire

$$5 \times 72 \text{ cm} = 360 \text{ cm}$$

$$\begin{array}{r} 72 \\ \times 5 \\ \hline 360 \end{array}$$

The length of the wire is 360 cm.

Answer: 360 cm

Solution to Question

94

Step 1 : Find the perimeter of the square

$$4 \times 15 \text{ m} = 60 \text{ m}$$

$$\begin{array}{r} 15 \\ \times 4 \\ \hline 60 \end{array}$$

Step 2 : Find the perimeter of the rectangle

$$18 \text{ m} + 5 \text{ m} + 18 \text{ m} + 5 \text{ m} = 46 \text{ m}$$

Step 3 : Find half of the length of the wire

$$60 \text{ m} + 46 \text{ m} = 106 \text{ m}$$

$$\begin{array}{r} 60 \\ + 46 \\ \hline 106 \end{array}$$

Step 4 : Find the length of the wire

$$2 \times 106 \text{ m} = 212 \text{ m}$$

$$\begin{array}{r} 106 \\ \times 2 \\ \hline 212 \end{array}$$

The length of the whole piece of wire is 212 m.

Answer: 212 m

Solution to Question

95

Step 1 : Find the area of the room

$$12 \text{ m} \times 9 \text{ m} = 108 \text{ m}^2$$

$$\begin{array}{r} 12 \\ \times 9 \\ \hline 108 \end{array}$$

Step 2 : Find the total cost of tiling the room

$$108 \text{ m}^2 \times \$6 = \$648$$

$$\begin{array}{r} 108 \\ \times 6 \\ \hline 648 \end{array}$$

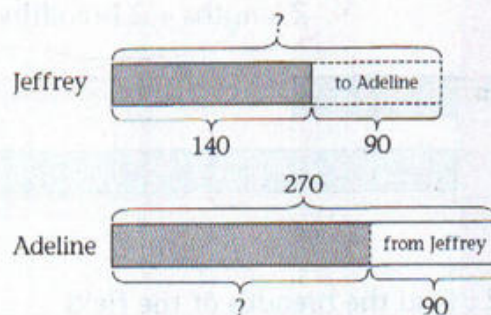
The total cost of tiling the room is \$648.

Answer: \$648

Solution to Question

96

Step 1 : Draw a model



Step 2 : Find the number of sweets Jeffrey had at first

$$140 + 90 = 230$$

$$\begin{array}{r} 140 \\ + 90 \\ \hline 230 \end{array}$$

(a) Jeffrey had 230 sweets at first.

Step 3 : Find the number of sweets Adeline had at first

$$270 - 90 = 180$$

$$\begin{array}{r} 270 \\ - 90 \\ \hline 180 \end{array}$$

Step 4 : Find the number of sweets Jeffrey had more than Adeline at first

$$230 - 180 = 50$$

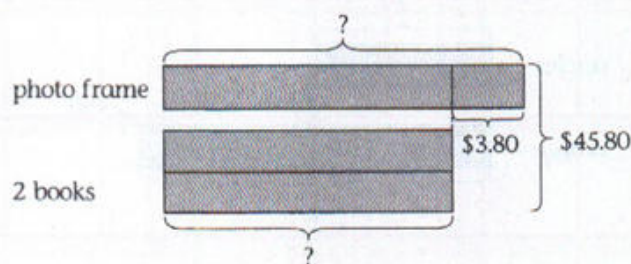
$$\begin{array}{r} 230 \\ - 180 \\ \hline 50 \end{array}$$

(b) Jeffrey had 50 sweets more than Adeline at first.

Answers: (a) 230 sweets

(b) 50 sweets

Step 1 : Draw a model



Step 2 : Find the value of 3 units

$$\$45.80 - \$3.80 = \$42$$

$$\begin{array}{r} 45.80 \\ - 3.80 \\ \hline 42.00 \end{array}$$

Step 3 : Find the cost of the book

$$3 \text{ units} \rightarrow \$42$$

$$1 \text{ unit} \rightarrow \$42 \div 3 = \$14$$

$$\begin{array}{r} 14 \\ 3 \overline{)42} \\ - 3 \downarrow \\ \hline 12 \\ - 12 \\ \hline 0 \end{array}$$

(a) The book was \$14.

Step 4 : Find the cost of the photo frame

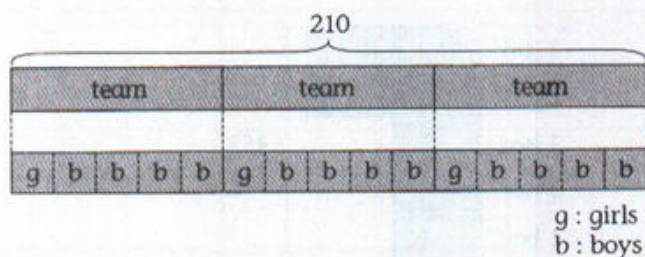
$$\$14 + \$3.80 = \$17.80$$

$$\begin{array}{r} 14.00 \\ + 3.80 \\ \hline 17.80 \end{array}$$

(b) The photo frame was \$17.80.

Answers: (a) \$14
(b) \$17.80

Step 1 : Draw a model



Step 2 : Find the number of children in a team

$$210 \div 3 = 70$$

$$\begin{array}{r} 70 \\ 3 \overline{)210} \\ - 21 \downarrow \\ \hline 00 \\ - 0 \\ \hline 0 \end{array}$$

Step 3 : Find the number of girls in a team

$$5 \text{ units} \rightarrow 70$$

$$1 \text{ unit} \rightarrow 70 \div 5 = 14$$

$$\begin{array}{r} 14 \\ 5 \overline{)70} \\ - 5 \downarrow \\ \hline 20 \\ - 20 \\ \hline 0 \end{array}$$

Step 4 : Find the total number of girls

$$3 \times 14 = 42$$

$$\begin{array}{r} 14 \\ \times 3 \\ \hline 42 \end{array}$$

(a) There were 42 girls altogether.

Step 5 : Find the total number of boys

$$210 - 42 = 168$$

$$\begin{array}{r} 210 \\ - 42 \\ \hline 168 \end{array}$$

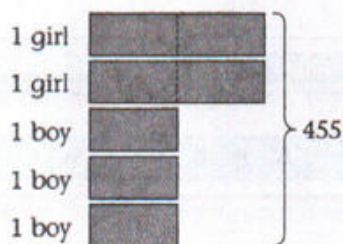
(b) There were 168 boys altogether.

Answers: (a) 42 girls
(b) 168 boys

Solution to Question

99

Step 1 : Draw a model



Step 2 : Find the number of cookies each boy will get

$$7 \text{ units} \rightarrow 455$$

$$1 \text{ unit} \rightarrow 455 \div 7 = 65$$

$$\begin{array}{r} 65 \\ 7 \overline{) 455} \\ \underline{- 42} \\ 35 \\ \underline{- 35} \\ 0 \end{array}$$

Step 3 : Find the total number of cookies the girls get

$$4 \text{ units} \rightarrow 4 \times 65 = 260$$

$$\begin{array}{r} 260 \\ 4 \times 65 \\ \hline \end{array}$$

(a) The girls get 260 cookies altogether.

Step 4 : Find the total number of cookies the boys get

$$3 \text{ units} \rightarrow 3 \times 65 = 195$$

$$\begin{array}{r} 195 \\ 3 \times 65 \\ \hline \end{array}$$

(b) The boys get 195 cookies altogether.

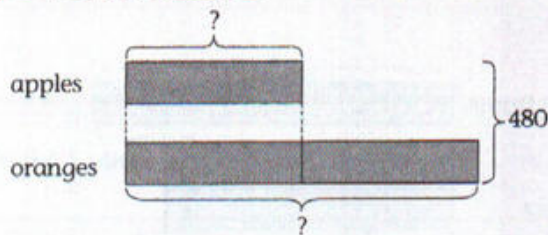
Answers: (a) 260 cookies

(b) 195 cookies

Solution to Question

100

Step 1 : Draw a model



Step 2 : Find the number of apples

$$3 \text{ units} \rightarrow 480$$

$$1 \text{ unit} \rightarrow 480 \div 3 = 160$$

$$\begin{array}{r} 160 \\ 3 \overline{) 480} \\ \underline{- 3} \\ 18 \\ \underline{- 18} \\ 0 \end{array}$$

(a) There were 160 apples.

Step 3 : Find the number of oranges

$$2 \text{ units} \rightarrow 2 \times 160 = 320$$

$$\begin{array}{r} 320 \\ 2 \times 160 \\ \hline \end{array}$$

(b) There were 320 oranges.

Answers: (a) 160 apples

(b) 320 oranges

Solution to Question

101

Step 1 : Find the area of the rectangular plot of land

$$45 \text{ m} \times 9 \text{ m} = 405 \text{ m}^2$$

$$\begin{array}{r} 405 \\ 45 \times 9 \\ \hline \end{array}$$

Step 2 : Find the area of the grass patch

$$8 \text{ m} \times 8 \text{ m} = 64 \text{ m}^2$$

Step 3 : Find the area of land not covered by the swimming pool and grass patch

$$405 \text{ m}^2 - 250 \text{ m}^2 = 155 \text{ m}^2$$

$$\begin{array}{r} 155 \\ 405 \\ \underline{- 250} \\ 155 \end{array}$$

$$155 \text{ m}^2 - 64 \text{ m}^2 = 91 \text{ m}^2$$

$$\begin{array}{r} 91 \\ 155 \\ \underline{- 64} \\ 91 \end{array}$$

91 m² of the land is not covered by the swimming pool and grass patch.

Answer: 91 m²